"Reinforcement Learning-Based Identification Of Failure Scenarios For Concept: Dynamic Risk Assessment Of AI-Controlled Robotic Systems (3D Simulation for Training & Testing Robots)"

PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT

FOR THE AWARD OF THE DEGREE OF

BACHELOR OF TECHNOLOGY

(Information Technology)

SUBMITTED BY

VISHNU VISHWAKARMA (22354)

GYANESHVAR (22328)

Under the supervision of

ER. AKHILESH KUMAR

ASSISTANT PROFESSOR

(I.T. Department)

डॉ. राममनोहर लोहिया अवध विश्वविद्यालय

ज्योतिषां ज्योतिरेकम्

July 2026

Institute of Engineering & Technology

Dr. Rammanohar Lohia Avadh University, Ayodhya (U.P.)

Session: 2022 – 2026

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CERTIFICATE

This certifies that **Vishnu Vishwakarma (22354)** and **Gyaneshvar (22328)** have conducted the work presented in this project report entitled **"REINFORCEMENT LEARNINGBASED IDENTIFICATION OF FAILURE SCENARIOS FOR CONCEPT: DYNAMIC RISK ASSESSMENT OF AI-CONTROLLED ROBOTIC SYSTEMS (3D Simulation for Training & Testing Robots)"** in the field of Information Technology at the Institute of Engineering & Technology**,** affiliated with Dr. R.M.L. Avadh University, Ayodhya, Uttar Pradesh**.** The project was carried out under the supervision of **Er. Akhilesh Kumar (Assistant Professor of IT Department).** This project represents the original work and studies conducted by the authors, and the contents do not form the basis for the award of any other degree to the candidates or anyone else from this or any other university/institution.

Er. Akhilesh Kumar	**Er. Rajesh Kumar Singh**
(Project Guide)	**(Head of Department)**
Assistant Professor	**Department of IT**
Department of IT	

CANDIDATE'S DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"REINFORCEMENT LEARNING-BASED IDENTIFICATION OF FAILURE SCENARIOS FOR CONCEPT: DYNAMIC RISK ASSESSMENT OF AI-CONTROLLED ROBOTIC SYSTEMS (3D Simulation for Training & Testing Robots)"** submitted by **Vishnu Vishwakarma (22354)** and **Gyaneshvar (22328)** in partial fulfillment of the requirements for the award of the degree of B.Tech (IT) submitted in the Department of I.T. at Institute of Engineering & Technology (IET), Dr. Ram Manohar Lohia Avadh University (RMLAU), Faizabad, Ayodhya (U.P.) is an authentic record of my own work carried out during a period of the B.Tech (IT)-Final Year (Semester-VII & Semester-VIII) under the supervision of **Er. Akhilesh Kumar.**

Vishnu Vishwakarma (22354)

Gyaneshvar (22328)

This is to certify that the above statement made by the candidate is correct to the best of my/our knowledge.

Signature of the Supervisor (s)

The B.Tech Viva-voce Examination of **Vishnu Vishwakarma (22354)** and **Gyaneshvar (22328)** has been held on......................and accepted.

Signature of Supervisor (s)	**Signature of Internal Examiner**
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Signature of H.O.D.	**Signature of External Examiner**
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ABSTRACT

AI-controlled robotic systems can pose significant risks to both human workers and the environment due to their complex, autonomous operations. Traditional risk assessment methods fall short in adequately describe sophisticated failure scenarios of systems controlled by black-box policies. These traditional methods are applied manually and do not considering frequent software updates. Therefore, novel methods that enable a dynamic risk assessment for AI-controlled robotic systems are required. In this paper, we present the concept of a new approach that enables the dynamic risk assessment of unknown or dynamically changing control policies. This includes AI-controlled robotic systems, especially those utilizing reinforcement learning. The new approach combines two key methods: (i) automated identification of failure modes and (ii) a reinforcement learning-based search for critical failure scenarios. Additionally, it incorporates an evaluation method that analyzes the detected sets of critical scenarios or events. This evaluation method creates dynamic high-level risk models and identifies vulnerabilities to specific failures.AI-controlled robotic systems pose a risk to human workers and the environment. Classical risk assessment methods cannot adequately describe such black box systems. Therefore, new methods for a dynamic risk assessment of such AI-controlled systems are required. In this paper, we introduce the concept of a new dynamic risk assessment approach for AI-controlled robotic systems. The approach pipelines five blocks: (i) a Data Logging that logs the data of the given simulation, (ii) a Skill Detection that automatically detects the executed skills with a deep learning technique, (iii) a Behavioral Analysis that creates the behavioral profile of the robotic systems, (iv) a Risk Model Generation that atomatically transforms the behavioral profile and risk data containing the failure probabilities of robotic hardware components into advanced hybrid risk models, and (v) Risk Model Solvers for the numerical evaluation of the generated hybrid risk models. Genetic Algorithms (G.A.s) are utilized to optimize fault injection parameters and automatically identify critical failure scenarios in AI-controlled robotic systems. This enables efficient exploration of fault combinations and thereby enhances/improving the effectiveness of dynamic risk assessment by discovering high-risk system behaviors. The project successfully demonstrates that modern technologies like AI and ML can be used to make traditional farming smarter, more efficient, and more sustainable. The system is designed to be low-cost, accessible through any web browser, and simple enough for farmers with minimal technical knowledge.

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Vishnu Vishwakarma (22354)

Gyaneshvar (22328)

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CHAPTER-01 : Introduction

Robotic systems can be classified as safety-critical due to their operation in close proximity to humans and frequent interactions with human operators. Such systems often incorporate moving components that could pose risks to human workers or handle hazardous substances, such as chemical mixtures. Robotic systems controlled by AI algorithms, especially reinforcement learning, have been a topic of interest for some time [1]. Recently, the application of reinforcement learning has experienced a significant increase in popularity [2]. Despite its growing adoption, traditional reinforcement learning lacks inherent safety guarantees. Dynamically changing control policies are essential for flexible production systems with high reconfigurability and repurpose-ability. They allow robotic systems to adapt to varying operational conditions and requirements, ensuring optimal performance. This adaptability enhances efficiency and enables rapid responses to changing production demands, making the manufacturing process more resilient and versatile. However, each new control policy can alter the behavior of the robotic systems, necessitating a renewed risk assessment. This underscores the need for a paradigm shift in risk assessment methodologies—from a static, one-time assessment prior to deployment to a dynamic risk assessment approach that continuously evaluates potential risks throughout the lifecycle of the system. The pivotal research question that emerges is: How can we dynamically assess the potential risks of robotic systems controlled by black-box policies? Robotic systems can be safety-critical because they are often deployed in close proximity to human workers. Moreover, they collaborate with human workers. They may contain moving parts that can pose risks to human workers or carry substances dangerous to humans, such as chemical mixtures, as shown in Figure 1. Robotic systems are increasingly being controlled by AI algorithms such as Reinforcement Learning [1]. This requires a shift of the paradigm of risk assessment towards a dynamic risk assessment instead of the classical approach when the risk assessment of the system is done only once before the operation. The research question to be answered is: How to automatically and dynamically estimate the potential risk of robotic systems controlled by black-box policies?

To efficiently explore a large search space of potential fault conditions, Genetic Algorithms (GAs) can be incorporated as an optimization technique. Genetic Algorithms mimic the process of natural evolution to identify optimal combinations of fault parameters, such as fault type, fault location, fault duration, and fault magnitude. By evolving candidate fault scenarios over multiple generations, GAs can efficiently discover critical and high-risk failure conditions that may not be detected through conventional testing approaches. The integration of Genetic Algorithms with Reinforcement Learning and Fault Injection techniques can significantly enhance the identification of hazardous robotic behaviors, thereby supporting dynamic risk assessment and improving the safety, reliability, and robustness of AI-controlled robotic systems.

Contribution: This paper introduces the concept of a novel approach for dynamic risk assessment of control policies that are either unknown or subject to dynamic changes, with a particular focus on AI-controlled robotic systems that employ reinforcement learning. The proposed method is illustrated in Figure 1. This innovative approach integrates two principal techniques: automated identification of failure modes and a reinforcement learning-based search for critical failure scenarios. Additionally, it includes an evaluation method that analyzes the detected sets of critical scenarios or events. This method constructs dynamic high-level risk models and identifies vulnerabilities to specific failures. The applicability of the proposed approach will be evaluated in a real setup with a Franka Emika Panda robot and reinforcement learning policies. This paper presents a concept of a new approach that enables the dynamic risk assessment of unknown or dynamic control policies including AI-controlled robotic systems. The proposed pipeline, shown in Figure 2, consists of five blocks:

- Data Logging consists of a hardware-interface integrated ROS logger. This block analyzes the in simulation executed ROS code. It logs time-series data such as the positions and velocities of each joint.
- Skill Detection block is a deep learning-based classifier. This classifier detects the executed skills and labels the time-series data with these skills e.g. move, pick, carry, etc.
- Behavioral Analysis block generates the behavioral profile of the given robotic systems. The behavioral profile contains information about the sequence of executed skills, active components for each skill, and properties of components such as velocity, execution time, intensity, etc. that influence reliability characteristics of the components.
- Risk Model Generation block transforms the behavioral profile and risk data, e.g. failure probabilities of robotic hardware components, into advanced hybrid risk models.
- Finally, the Risk Model Solver quantifies the generated risk models.

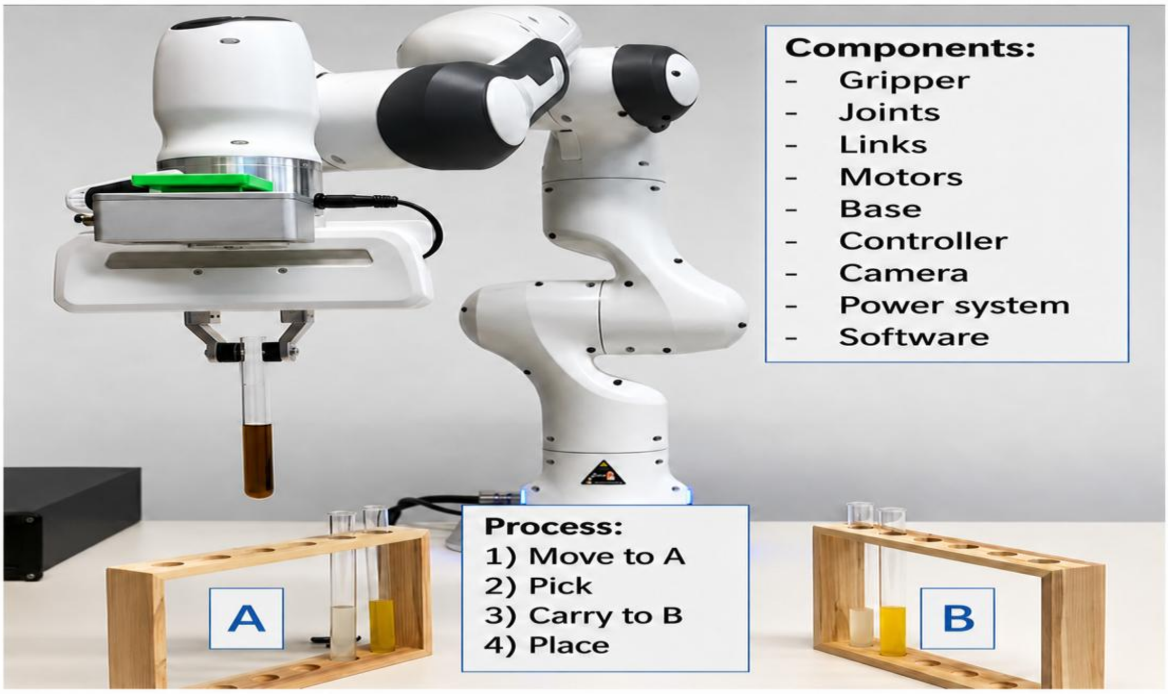


Figure 1.1. Robotic manipulator which has been trained to grasp a test tube at position A and place it to position B.

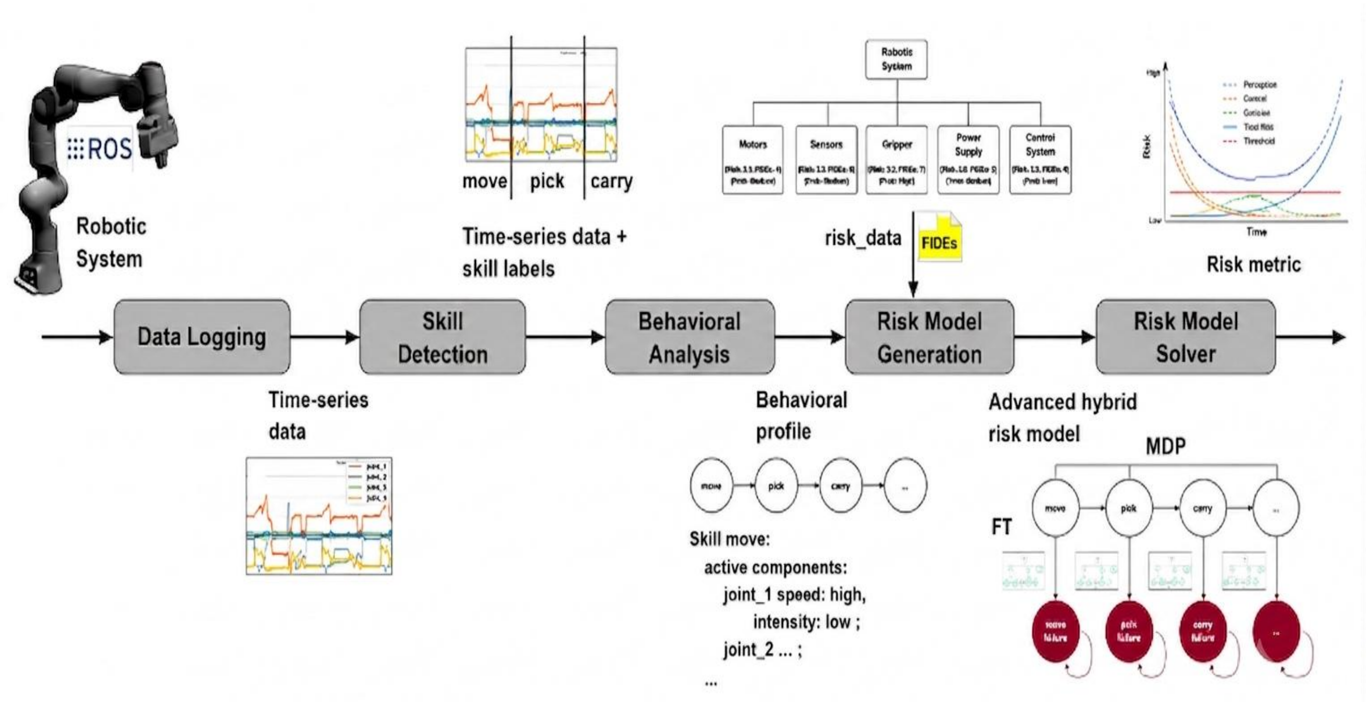


Figure 1.2. The proposed approach to the dynamic risk assessment of AI-controlled robotic systems.

CHAPTER-02 : Problem Statement

2.1 Background

The rapid adoption of Artificial Intelligence (AI), particularly Reinforcement Learning (RL), has significantly transformed modern robotic systems by enabling autonomous decisionmaking and adaptive behavior. These intelligent robotic systems are increasingly deployed in safety-critical environments such as industrial automation, smart manufacturing, healthcare, laboratories, logistics, and human-robot collaborative workspaces. Although Reinforcement Learning enables robots to learn optimal control policies through interaction with their environment, the learned policies often behave as black-box models, making their internal decision-making process difficult to understand, interpret, and verify.

2.2 Limitations of Traditional Risk Assessment

Traditional risk assessment and safety analysis methods were primarily designed for deterministic robotic systems whose behavior remains fixed after deployment. These methods assume that system behavior does not change over time and therefore perform risk analysis only once during the design phase. However, this assumption is no longer valid for AIcontrolled robotic systems. Reinforcement Learning agents continuously adapt their policies based on environmental interactions, software updates, retraining, and changing operational conditions. Consequently, the behavior of robotic systems evolves dynamically, making conventional static risk assessment techniques inadequate for identifying newly emerging hazards.

2.3 Challenges in Identifying Critical Failure Scenarios

A major limitation of existing approaches is their inability to automatically identify critical failure scenarios before system deployment. Conventional testing methods rely on manually designed fault cases and predefined hazardous situations. These approaches fail to explore the enormous search space of possible fault combinations involving sensor failures, actuator malfunctions, communication delays, software faults, environmental uncertainties, and unexpected human interactions. As a result, many dangerous failure scenarios remain undiscovered until real-world deployment, where failures may cause equipment damage, production losses, environmental hazards, or serious injuries to human workers.

2.4 Limitations of Existing Dynamic Risk Assessment Methods

Existing risk assessment techniques also struggle to analyze robotic systems controlled by unknown or black-box AI policies because the internal decision-making logic cannot be directly inspected. Without understanding how failures propagate through the learned policy, it becomes extremely difficult to estimate the probability, severity, and consequences of hazardous events. Furthermore, current methodologies lack mechanisms to continuously update risk models whenever the robot learns new behaviors or receives modified control policies.

2.5 Challenges in Simulation and Fault Injection

Simulation-based testing has become a widely adopted solution for evaluating robotic systems before real-world deployment. However, traditional simulation methods execute only predefined scenarios and cannot intelligently search for the most hazardous combinations of faults. Although Reinforcement Learning-based exploration can automatically discover unsafe system behaviors, it often suffers from inefficient exploration, sparse rewards, slow convergence, and difficulty in identifying rare but catastrophic failure events.

Similarly, fault injection techniques improve system robustness by introducing artificial faults such as sensor noise, communication failures, packet loss, actuator bias, bit flips, and joint malfunctions. However, manually selecting fault parameters—including fault type, location, timing, duration, and intensity—is computationally expensive and does not guarantee the discovery of the most critical failure scenarios.

2.6 Research Gap

Current dynamic risk assessment frameworks primarily focus on hardware failures and provide limited support for integrating software behavior, AI decision-making, autonomous learning, and intelligent failure analysis into a unified framework. Moreover, there is no comprehensive approach capable of combining simulation, fault injection, behavioral analysis, Reinforcement Learning, Genetic Algorithms, Large Language Models (LLMs), and dynamic risk modeling for continuous safety evaluation.

2.7 Need for the Proposed System

To overcome these limitations, there is a need for an intelligent Dynamic Risk Assessment framework capable of continuously monitoring AI-controlled robotic systems, automatically identifying potential failure modes, exploring hazardous failure scenarios, and quantitatively evaluating associated risks throughout the system lifecycle.

The proposed framework should integrate:

- Reinforcement Learning for intelligent exploration of hazardous scenarios.
- Fault Injection techniques for realistic fault simulation.
- Deep Learning for robot skill detection and behavioral analysis.
- Genetic Algorithms for optimizing fault injection parameters.
- Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) for automated failure mode identification.
- Dynamic Risk Models such as Fault Trees, Bayesian Networks, Markov Chains, and OpenPRA for quantitative risk evaluation.

2.8 Problem Definition

The primary problem addressed in this project is the absence of a comprehensive, adaptive, and automated framework capable of dynamically assessing risks in AI-controlled robotic systems operating under Reinforcement Learning-based black-box control policies. Existing approaches are unable to continuously identify emerging hazards, automatically discover critical failure scenarios, and update risk models according to changing robotic behaviors.

To address this research gap, this project proposes a Reinforcement Learning-based Dynamic Risk Assessment framework that integrates Data Logging, Skill Detection, Behavioral Analysis, Fault Injection, Failure Mode Identification, Reinforcement Learning-based Failure Scenario Analysis, Genetic Algorithm optimization, and Dynamic Risk Modeling. The proposed framework aims to improve the safety, reliability, robustness, and trustworthiness of next-generation AI-controlled robotic systems operating in safety-critical environments.

CHAPTER-03

3.1 Objectives Of The Project

The primary objective of this project is to develop an intelligent framework for identifying failure scenarios in AI-controlled robotic systems using Reinforcement Learning (RL). The proposed system focuses on improving the safety and reliability of autonomous robots by continuously monitoring their behavior and performing Dynamic Risk Assessment (DRA).

Another objective is to design a simulation-based environment using the Robot Operating System (ROS) and Gazebo, where different robotic tasks can be executed safely without using physical hardware. The simulation environment allows the robot to perform operations such as object picking, movement, carrying, and placement while generating operational data for analysis.

The project also aims to implement automated modules including Data Logging, Skill Detection, Behavioral Analysis, Risk Model Generation, Failure Mode Identification (RiskGPT), and Reinforcement Learning-based Failure Scenario Analysis. These modules work together to identify potential risks and estimate the probability of failures during robot operation.

Another important objective is to reduce the dependency on manual safety analysis by introducing AI-based automation in hazard identification and risk evaluation. The proposed framework also investigates the use of Genetic Algorithms to improve the efficiency of fault exploration and identify critical failure scenarios more effectively.

Overall, the project aims to develop a scalable and intelligent Dynamic Risk Assessment framework that enhances the safety, reliability, and robustness of AI-controlled robotic systems operating in complex and dynamic environments.

3.2 Scope of the Project

The scope of this project is focused on developing a Dynamic Risk Assessment framework for AI-controlled robotic systems using Reinforcement Learning. The framework is implemented in a simulated environment using ROS and Gazebo, allowing robotic operations to be tested without the risk of damaging physical hardware.

The project includes modules for data collection, skill detection, behavioral analysis, risk model generation, failure mode identification, and reinforcement learning-based failure scenario analysis. These modules work together to monitor robot behavior, identify potential hazards, and evaluate system safety continuously.

The proposed framework can be applied in various domains such as industrial automation, warehouse robotics, healthcare, autonomous laboratories, and smart manufacturing. Although the current implementation is performed in a simulation environment, the methodology can be extended to real robotic systems with suitable hardware integration.

The project is limited to AI-controlled robotic manipulators operating in simulated environments and does not include deployment on physical industrial robots. However, the proposed framework provides a strong foundation for future research on safe and reliable autonomous robotic systems.

3.3 Significance of the Project

Artificial Intelligence and Reinforcement Learning have significantly improved the capabilities of modern robotic systems by enabling them to perform complex tasks autonomously. However, these systems continuously update their decision-making policies during learning, making traditional static risk assessment methods less effective. Therefore, ensuring the safety and reliability of AI-controlled robots has become an important research area.

The significance of this project lies in the development of a Dynamic Risk Assessment (DRA) framework that continuously monitors robotic behavior and identifies potential failure scenarios during task execution. Unlike conventional approaches, the proposed framework evaluates system safety throughout the robot's operational lifecycle rather than only before deployment.

The project integrates Reinforcement Learning, ROS, Gazebo simulation, RiskGPT, and probabilistic risk analysis to create an intelligent safety assessment framework. This integration helps in automatically identifying possible hazards, analyzing failure modes, and improving the overall reliability of autonomous robotic systems.

The proposed methodology is useful for applications such as industrial automation, warehouse management, healthcare robotics, research laboratories, and smart manufacturing, where robotic safety is a critical requirement. Overall, this project contributes towards developing safe, reliable, and trustworthy AI-controlled robotic systems for future intelligent industries.

3.4 Organization of the Report

This project report has been organized into different chapters to provide a systematic understanding of the proposed research work.

Chapter 1 introduces the project by discussing the background, problem statement, objectives, scope, significance, and motivation of the research.

Chapter 2 presents the literature review, including recent research papers, existing technologies, private startups, research gaps, and the need for the proposed Dynamic Risk Assessment framework.

Chapter 3 explains the complete design and implementation of the proposed system. It includes system analysis, architecture, implementation details, testing methodology, performance evaluation, comparison with existing systems, project achievements, applications, and future directions.

Finally, the report concludes with the overall findings of the project and provides recommendations for future improvements. This organization allows readers to understand the project in a logical and structured manner while maintaining continuity throughout the report.

3.5 Private Startups and Projects

The rapid growth of Artificial Intelligence (AI) and robotics has encouraged several private companies and startups to develop intelligent robotic systems capable of performing complex tasks autonomously. These organizations are focusing on reinforcement learning, computer vision, simulation technologies, and intelligent automation to improve robotic performance in industrial and real-world environments. Their research has significantly contributed to the advancement of autonomous robotics and has inspired the development of safer and more reliable robotic systems.

One of the leading organizations in this field is **Boston Dynamics**, which develops highly advanced mobile robots such as Spot and Atlas. These robots are capable of autonomous navigation, object handling, and operation in difficult environments. Their work demonstrates the practical application of AI in robotics and highlights the importance of safety during autonomous operation.

Another important organization is **NVIDIA**, which provides AI computing platforms and robotic simulation environments such as Isaac Sim. These tools enable researchers to train and test reinforcement learning algorithms in realistic virtual environments before deploying them on actual robots. Simulation-based development greatly reduces cost, improves efficiency, and minimizes operational risks.

Covariant is another AI startup that develops intelligent warehouse robots using deep learning and reinforcement learning. Their robotic systems automatically identify, pick, and sort objects of different shapes and sizes, improving warehouse automation and logistics operations.

Intrinsic, a robotics software company under Alphabet, focuses on simplifying robot programming using AI technologies. The company develops intelligent software solutions that allow industrial robots to learn new tasks with minimal programming effort.

Similarly, companies such as **Figure AI** and **Sanctuary AI** are developing humanoid robots capable of performing human-like tasks in industrial and commercial environments. These organizations emphasize intelligent decision-making, autonomous learning, and safe humanrobot collaboration.

The work carried out by these startups demonstrates the increasing demand for intelligent robotic systems. However, most existing solutions primarily focus on improving robot performance and automation, whereas continuous dynamic risk assessment remains a relatively unexplored area. The proposed project addresses this limitation by integrating reinforcement learning, behavioral analysis, RiskGPT, and probabilistic risk assessment into a unified safety evaluation framework.

3.6 Related Research Papers

Several researchers have proposed different techniques to improve the safety and reliability of AI-controlled robotic systems. Most existing studies focus on reinforcement learning, robotic behavior analysis, fault diagnosis, and probabilistic risk assessment.

Recent research on Dynamic Risk Assessment suggests that continuous monitoring of robotic behavior provides more accurate safety evaluation than traditional static methods. Researchers have developed various probabilistic models such as Fault Tree Analysis (FTA), Bayesian Networks, Reliability Block Diagrams (RBD), and Markov Models to estimate system reliability and failure probability.

Many studies have also explored the application of Reinforcement Learning in robotics. These algorithms enable robots to learn optimal actions through interaction with the environment instead of relying on manually programmed rules. Although RL improves robot adaptability, continuously changing control policies introduce new safety challenges that require dynamic monitoring.

Large Language Models (LLMs) have recently been applied to engineering applications for automated documentation, code analysis, and fault identification. The concept of RiskGPT extends these capabilities by automatically identifying failure modes, their causes, and possible mitigation strategies for robotic systems.

Researchers have further investigated robotic simulation platforms such as ROS and Gazebo for testing intelligent control algorithms. Simulation-based experimentation allows developers to validate robotic behavior under different operating conditions without risking physical hardware.

Although previous research has made significant progress in AI-based robotics, most studies focus on individual aspects such as robot learning, simulation, or fault analysis. Very few studies integrate reinforcement learning, behavioral analysis, automated failure mode identification, and probabilistic risk modeling into a single framework. This research attempts to bridge that gap by developing a comprehensive Dynamic Risk Assessment framework capable of continuously evaluating AI-controlled robotic systems.

3.7 Gaps Identified

A detailed study of existing research papers and industrial robotic systems reveals that although significant progress has been made in Artificial Intelligence and Reinforcement Learning, several research gaps still exist in the field of robotic safety and Dynamic Risk Assessment.

Most existing robotic systems focus primarily on improving task performance, learning efficiency, and automation. Safety evaluation is generally performed before deployment using static risk assessment methods, which are unable to adapt when the robot learns new behaviors during operation.

Another major gap is that existing approaches usually analyze only individual system components such as sensors, controllers, or actuators. Very few frameworks provide a complete end-to-end analysis by combining data collection, behavioral analysis, failure mode identification, probabilistic risk assessment, and reinforcement learning within a single system.

Current fault analysis techniques also rely heavily on manual identification of hazards and expert knowledge. This process is time-consuming, expensive, and may overlook rare or unexpected failure scenarios. Moreover, conventional testing methods cannot efficiently explore all possible fault combinations in complex robotic environments.

Although simulation platforms such as ROS and Gazebo are widely used for robot development, they are mainly employed for algorithm testing rather than continuous safety monitoring and dynamic risk evaluation. Similarly, recent AI technologies such as Large Language Models have shown promising results in software engineering, but their application in automated robotic failure identification is still limited.

Therefore, there is a clear need for an intelligent framework capable of continuously monitoring robotic behavior, automatically identifying hazards, generating probabilistic risk models, and discovering failure scenarios using Artificial Intelligence. These identified gaps form the motivation for the proposed research.

3.8 How Our Project Addresses These Gaps

The proposed project has been designed specifically to overcome the limitations identified in existing research and industrial robotic systems. Instead of relying on traditional static safety assessment methods, the proposed framework performs Dynamic Risk Assessment by continuously monitoring the robot throughout its operation.

The framework integrates multiple intelligent modules, including Data Logging, Skill Detection, Behavioral Analysis, Risk Model Generation, RiskGPT, and Reinforcement Learning-based Failure Scenario Analysis. These modules work together to automatically collect operational data, analyze robotic behavior, identify possible failure modes, and estimate the probability of potential risks without requiring extensive manual intervention.

Unlike conventional approaches that depend heavily on expert knowledge for hazard identification, the proposed system utilizes RiskGPT to automatically generate possible failure modes and mitigation strategies. Furthermore, the Reinforcement Learning agent intelligently explores different fault conditions to discover hazardous operating scenarios that may not be identified through traditional testing methods.

The use of ROS and Gazebo provides a safe simulation environment where robotic tasks can be executed repeatedly under different operating conditions. This allows comprehensive testing of AI-controlled robotic systems without risking physical hardware or human operators.

By combining Reinforcement Learning, probabilistic risk assessment, behavioral analysis, and intelligent failure identification into a unified framework, the proposed project provides a more reliable, scalable, and efficient solution for improving the safety and trustworthiness of autonomous robotic systems. The methodology can also be extended to future applications in smart manufacturing, healthcare robotics, warehouse automation, autonomous laboratories, and other AI-driven industrial environments.

3.9 System Analysis and Design

System Analysis and Design is an important phase in the development of the proposed Dynamic Risk Assessment framework. It involves understanding the system requirements, identifying functional components, and designing an efficient architecture for safe and reliable robotic operation. The analysis focuses on how AI-controlled robotic systems perform different tasks, collect operational data, identify possible risks, and generate safety assessments during execution.

The proposed system consists of multiple interconnected modules including Data Logging, Skill Detection, Behavioral Analysis, Risk Model Generation, RiskGPT, Failure Scenario Analysis, and Risk Model Solver. Each module performs a specific function while exchanging information with other modules to ensure smooth operation. This modular design improves scalability, simplifies maintenance, and allows future enhancements without affecting the overall system performance.

The system is implemented using the Robot Operating System (ROS) and the Gazebo simulation environment. Reinforcement Learning algorithms are used to train the robotic agent, while probabilistic risk assessment techniques evaluate the safety and reliability of the system. Overall, the proposed design provides an efficient platform for continuous monitoring and intelligent risk analysis of autonomous robotic systems.

3.10 System Architecture

The proposed system follows a modular architecture in which all components work together to perform Dynamic Risk Assessment of AI-controlled robotic systems. The process begins with the robotic simulation environment developed using ROS and Gazebo, where the robot performs different manipulation tasks such as movement, object picking, carrying, and placement.

During task execution, the Data Logging module continuously records sensor values, joint positions, velocities, actuator efforts, and controller information. These data are then processed by the Skill Detection module to identify the robot's current activity. The Behavioral Analysis module studies the execution pattern and generates behavioral profiles representing the robot's operational characteristics.

The generated behavioral information is forwarded to the Risk Model Generator, which automatically creates probabilistic risk models. Simultaneously, RiskGPT identifies possible failure modes and recommends suitable mitigation strategies. The Reinforcement Learningbased Failure Scenario Analyzer explores different fault combinations to identify hazardous operating conditions. Finally, the Risk Model Solver calculates failure probabilities and provides a comprehensive risk assessment report for the robotic system.

This modular architecture ensures efficient communication between all components while supporting continuous monitoring, automated analysis, and intelligent decision-making.

3.11 System Design Elements

The proposed framework is composed of several design elements that work together to improve the safety and reliability of AI-controlled robotic systems.

Simulation Environment: The system uses ROS and Gazebo to simulate robotic operations under different environmental conditions. Simulation enables safe testing without using expensive physical robots.

Data Logging Module: This module records all operational information generated during robot execution, including sensor readings, joint movements, actuator efforts, and controller outputs.

Skill Detection Module: It analyzes the collected data and identifies individual robotic actions such as movement, grasping, carrying, and object placement.

Behavioral Analysis Module: This component studies the sequence of robotic skills and evaluates system behavior to identify abnormal execution patterns.

Risk Model Generator: It automatically converts behavioral information into probabilistic models for estimating system reliability and failure probability.

RiskGPT Module: This AI-based module identifies possible failure modes, analyzes their causes, predicts their impact, and suggests mitigation techniques.

Failure Scenario Analyzer: Using Reinforcement Learning, this module explores different fault conditions and discovers hazardous scenarios that may affect robotic performance.

Risk Model Solver: It performs quantitative analysis of generated risk models and produces the final safety evaluation report.

The modular structure improves flexibility, maintainability, and future scalability of the proposed framework.

3.12 Security and Privacy Design

Security and privacy are essential aspects of intelligent robotic systems, especially when robots continuously exchange operational data during autonomous execution. The proposed framework incorporates several security measures to ensure the integrity and reliability of collected information.

Operational data generated during robotic tasks are securely stored and accessed only by authorized system modules. The framework prevents unauthorized modification of sensor readings, behavioral data, and generated risk models. Proper authentication mechanisms can be implemented to control user access and protect sensitive information.

The use of simulation environments further enhances system security by allowing fault testing and reinforcement learning experiments without exposing physical robots or human operators to potential hazards. In future implementations, secure communication protocols, encrypted data transmission, and cloud-based authentication mechanisms can be integrated to strengthen cybersecurity in industrial robotic applications.

Overall, the proposed design ensures that safety assessment is performed without compromising data security or user privacy.

3.13 Design Considerations

Several important factors were considered during the design of the proposed Dynamic Risk Assessment framework. The first consideration was modularity, allowing each system component to operate independently while maintaining efficient communication with other modules. This approach simplifies debugging, maintenance, and future upgrades.

Scalability was another key consideration. The framework is designed so that additional robotic platforms, sensors, AI algorithms, or safety modules can be integrated with minimal modification to the existing architecture.

Reliability and accuracy were also considered during system development. Automated data collection, behavioral analysis, and probabilistic risk modeling improve the consistency of safety evaluation while reducing human errors. The use of reinforcement learning enables intelligent exploration of fault conditions that may not be discovered through conventional testing.

Finally, usability and flexibility were considered to ensure that the framework can be adapted for different robotic applications including industrial automation, healthcare robotics, warehouse management, research laboratories, and smart manufacturing systems. These design considerations make the proposed system efficient, reliable, and suitable for future AI-driven robotic environments.

3.14 Implementation Introduction

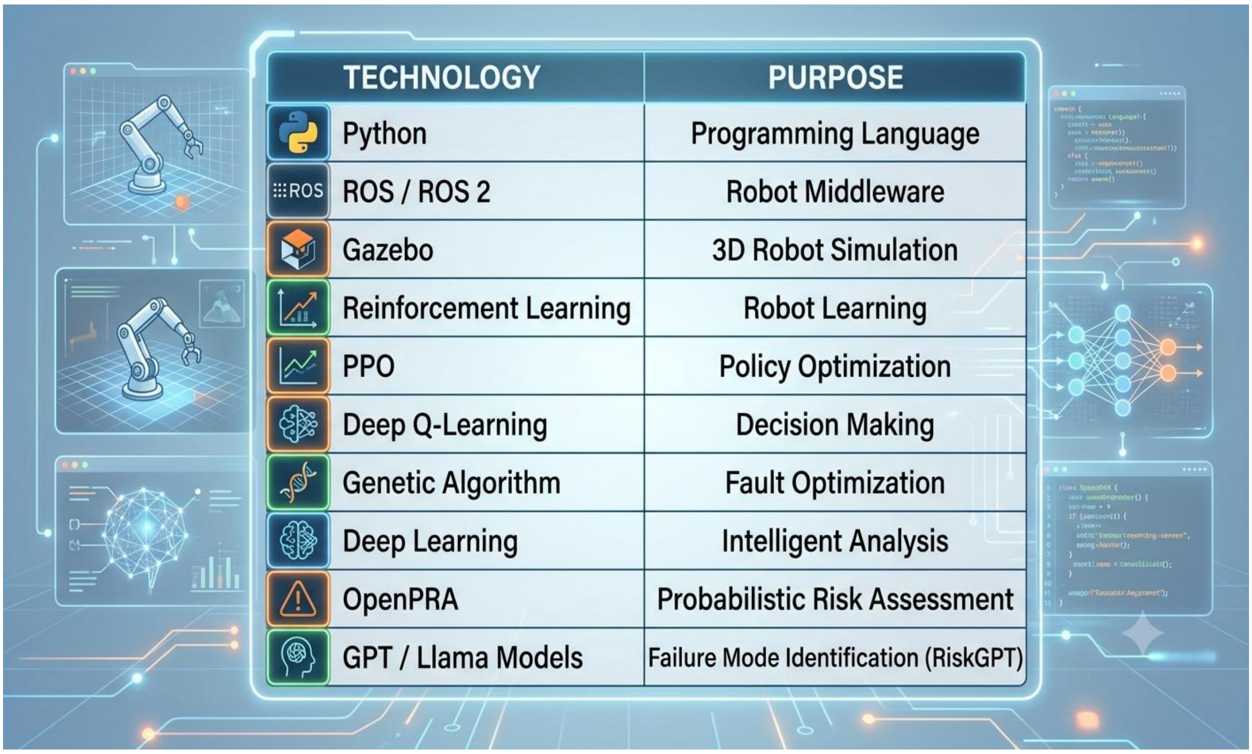
The implementation phase focuses on converting the proposed Dynamic Risk Assessment framework into a functional system. The framework integrates Artificial Intelligence, Reinforcement Learning, Robot Operating System (ROS), Gazebo simulation, and probabilistic risk assessment techniques to evaluate the safety of AI-controlled robotic systems. Each module was implemented independently and later integrated to achieve continuous monitoring, automated failure identification, and dynamic risk analysis. A simulation-based approach was adopted to ensure safe testing without the need for physical robotic hardware.

3.15 Development Environment

The proposed system was developed using a software-based simulation environment to ensure flexibility, safety, and cost-effectiveness. The Robot Operating System (ROS) was used as the middleware for communication between different modules, while Gazebo served as the 3D simulation platform for robotic task execution.

Python was used as the primary programming language because of its simplicity and extensive AI libraries. Reinforcement Learning algorithms were implemented for intelligent decisionmaking, while RiskGPT was utilized for automated failure mode identification. Ubuntu Linux provided a stable development platform for ROS-based applications. The combination of these technologies enabled efficient development, testing, and validation of the proposed framework.

3.15.1. Technologies Used



TECHNOLOGY	PURPOSE
Python	Programming Language
ROS / ROS 2	Robot Middleware
Gazebo	3D Robot Simulation
Reinforcement Learning	Robot Learning
PPO	Policy Optimization
Deep Q-Learning	Decision Making
Genetic Algorithm	Fault Optimization
Deep Learning	Intelligent Analysis
OpenPRA	Probabilistic Risk Assessment
GPT / Llama Models	Failure Mode Identification (RiskGPT)

Used Keywords:

1. Dynamic Risk Assessment,
2. Hybrid Risk Models,
3. Fault Injection,
4. M2M (Model-to-Model) Transformation,
5. ROS (Robot Operating System),
6. AI-Controlled Robotic Systems,
7. Genetic Algorithms (G.A.s),
8. Deep Learning (DL),
9. Large Language Models (LLMs)
10. & Reinforcement Learning (RL)

3.16 Implementation of Modules

The proposed framework consists of several interconnected modules that perform different functions during risk assessment.

Data Logging Module: Collects operational information such as joint positions, velocities, sensor readings, and actuator efforts during robot execution.

Skill Detection Module: Identifies the robot's current task by analyzing recorded operational data.

Behavioral Analysis Module: Studies execution patterns and generates behavioral profiles for different robotic skills.

Risk Model Generator: Converts behavioral information into probabilistic models for quantitative risk evaluation.

RiskGPT Module: Automatically identifies failure modes, analyzes possible causes, and suggests mitigation strategies using AI techniques.

Failure Scenario Analyzer: Uses Reinforcement Learning to generate and evaluate different fault conditions in the simulation environment.

Risk Model Solver: Calculates failure probabilities and produces the final Dynamic Risk Assessment report.

The integration of these modules enables continuous monitoring and intelligent safety evaluation throughout robotic operation.

3.17 Frontend & Backend Implementation

The proposed system follows a client-server architecture consisting of frontend and backend components. The frontend provides an interface for configuring simulation parameters, monitoring robot activities, and viewing generated results. It allows users to visualize robot behavior, analyze detected failures, and examine risk assessment reports.

The backend performs all computational tasks, including simulation control, data processing, reinforcement learning, behavioral analysis, probabilistic calculations, and report generation. Communication between the frontend and backend is managed through ROS nodes and service communication, ensuring efficient data exchange and real-time system performance.

3.18 Database Design

The database is responsible for storing operational data generated during robotic execution. It maintains information related to sensor readings, robot states, detected skills, behavioral profiles, failure modes, generated risk models, and assessment results.

The stored information allows repeated analysis, comparison of experimental results, and future model improvements. The database structure is designed to support efficient retrieval of information while maintaining consistency and reliability of recorded data. Proper organization of experimental data also facilitates performance evaluation and validation of the proposed framework.

3.19 Testing During Implementation

Testing was performed continuously during the implementation phase to ensure the correct functioning of each module. Individual modules were initially tested independently to verify their outputs before integrating them into the complete system.

Different robotic tasks were executed within the Gazebo simulation environment under normal and faulty operating conditions. The generated outputs were compared with expected results to verify the accuracy of data collection, behavioral analysis, risk model generation, and failure identification. Continuous testing throughout development helped identify implementation errors at an early stage and improved the overall reliability of the proposed framework.

3.20 Integration of Modules

After successful implementation of individual modules, all components were integrated into a unified Dynamic Risk Assessment framework. The Data Logging module supplied operational data to the Skill Detection and Behavioral Analysis modules. The generated behavioral information was then forwarded to the Risk Model Generator and RiskGPT for safety evaluation.

The Reinforcement Learning-based Failure Scenario Analyzer interacted with the simulation environment to identify hazardous operating conditions. Finally, the Risk Model Solver calculated failure probabilities and generated comprehensive safety reports. The successful integration of all modules enabled automated and continuous risk assessment throughout the robotic task execution process.

3.21 Deployment Plan

The proposed framework is primarily designed for deployment within a ROS-Gazebo simulation environment. Before real-world deployment, all robotic tasks and safety evaluations are validated through simulation to minimize operational risks.

For industrial implementation, the framework can be integrated with actual robotic manipulators by connecting real sensor data and controller outputs to the developed modules. Future deployment may also include cloud-based monitoring systems, remote supervision, automated report generation, and continuous safety assessment in industrial environments. This phased deployment strategy ensures reliable system validation while reducing implementation risks and operational costs.

3.22 Testing Strategy

A systematic testing strategy was adopted to evaluate the functionality, reliability, and safety of the proposed Dynamic Risk Assessment framework. Initially, each module was tested individually using unit testing to verify its functionality. After successful verification, integration testing was performed to ensure smooth communication among all modules. Finally, system testing was conducted by executing complete robotic tasks in the ROS-Gazebo simulation environment. The testing process focused on validating data collection, behavioral analysis, failure mode identification, risk model generation, and reinforcement learning-based fault analysis under different operating conditions.

3.23 Testing Environment

The proposed framework was tested in a simulated environment using the Robot Operating System (ROS) and Gazebo. Ubuntu Linux served as the development platform, while Python was used for implementing various AI and reinforcement learning algorithms. The simulation environment allowed multiple robotic tasks to be executed safely without requiring physical robotic hardware.

Different fault scenarios such as sensor failures, actuator malfunctions, and communication delays were introduced to evaluate the effectiveness of the Dynamic Risk Assessment framework. The simulation environment provided a flexible and cost-effective platform for validating the proposed methodology.

3.24 Test Cases

Several test cases were designed to verify the performance of different system modules. The major test cases included:

Test Case	Expected Result	Status
Robot Initialization	Robot starts successfully	Passed
Data Logging	Operational data recorded correctly	Passed
Skill Detection	Robot actions identified accurately	Passed
Behavioral Analysis	Behavioral profile generated	Passed
RiskGPT Analysis	Failure modes identified	Passed
Risk Model Generation	Probabilistic model created	Passed
Failure Scenario Analysis	Fault scenarios detected	Passed
Final Risk Assessment	Risk report generated successfully	Passed

The successful execution of these test cases confirmed the correct functioning of the proposed framework.

3.25 Results and Accuracy

Experimental results demonstrated that the proposed framework successfully monitored robotic behavior and generated Dynamic Risk Assessment reports during task execution. The Data Logging module accurately collected operational information, while the Skill Detection and Behavioral Analysis modules correctly identified robotic activities.

RiskGPT effectively generated possible failure modes, and the Reinforcement Learning agent successfully discovered several hazardous operating conditions within the simulation environment. The generated probabilistic risk models provided reliable estimates of system safety and failure probability. Overall, the experimental results indicate that the proposed framework improves the effectiveness of safety assessment for AI-controlled robotic systems.

3.26 Performance Testing

Performance testing was conducted to evaluate the efficiency of the proposed system under different operating conditions. Parameters such as execution time, response time, processing speed, data handling capability, and simulation stability were analyzed.

The framework maintained stable performance while executing multiple robotic tasks and processing large volumes of operational data. Automated risk model generation and failure identification were completed within acceptable computational time, demonstrating that the framework is suitable for continuous monitoring of AI-controlled robotic systems.

3.27 Limitations Found

Although the proposed framework achieved its objectives successfully, several limitations were identified during testing. The current implementation is limited to simulation-based evaluation and has not been deployed on physical industrial robots. The accuracy of risk assessment depends on the quality of simulation data and reinforcement learning training.

In addition, complex robotic environments with multiple interacting robots may require greater computational resources and longer training times. Future work can focus on improving computational efficiency and validating the framework using real-world robotic platforms.

3.28 Testing Tools Used

The proposed framework was tested using several software tools and development platforms.

- **Robot Operating System (ROS) –** Robot communication and control.
- **Gazebo –** 3D robotic simulation.
- **Python –** Algorithm implementation.
- **Reinforcement Learning Libraries –** Intelligent fault exploration.
- **RiskGPT –** Automated failure mode identification.
- **Ubuntu Linux –** Development and testing platform.

These tools provided a reliable environment for implementing, testing, and validating the proposed Dynamic Risk Assessment framework.

3.29 Achievements

The major achievements of this project include the successful development of an intelligent Dynamic Risk Assessment framework capable of monitoring AI-controlled robotic systems continuously.

The project successfully integrated ROS, Gazebo, Reinforcement Learning, RiskGPT, behavioral analysis, probabilistic risk assessment, and automated failure scenario identification into a unified framework. The obtained results demonstrate the feasibility of AI-driven safety assessment for autonomous robotic systems.

3.30 Applications of the Project

The proposed framework can be applied in several domains requiring intelligent robotic safety assessment, including:

- Smart Manufacturing Industries

- Warehouse Automation
- Healthcare and Medical Robotics
- Autonomous Research Laboratories
- Industrial Robotic Manipulators
- Agriculture and Smart Farming
- Defense and Surveillance Robotics
- Space Exploration Robotics
- Autonomous Mobile Robots
- Industry 5.0 Smart Factories

These applications demonstrate the broad industrial relevance and future potential of the proposed Dynamic Risk Assessment framework.

3.31 Final Thoughts

Artificial Intelligence is transforming the future of robotics by enabling autonomous decisionmaking and intelligent task execution. However, increasing autonomy also demands advanced safety mechanisms capable of identifying risks during system operation. The proposed
Dynamic Risk Assessment framework addresses this challenge by integrating Reinforcement Learning, behavioral analysis, RiskGPT, and probabilistic risk assessment into a unified intelligent system.

The successful implementation and evaluation of the proposed methodology demonstrate its potential to improve the safety, reliability, and trustworthiness of AI-controlled robotic systems. With further research and real-world deployment, this framework can contribute significantly to the development of next-generation autonomous robotic technologies for industrial, healthcare, and research applications.

CHAPTER-04 : STATE OF THE ART/Literature Survey/Project Idea

This section discusses three relevant groups of methods, which are crucial for explaining the novelty of the proposed concept. First, dynamic reliability and risk assessment methods for robotic systems. Second, large language models for risk assessment. Third, fault injection methods.

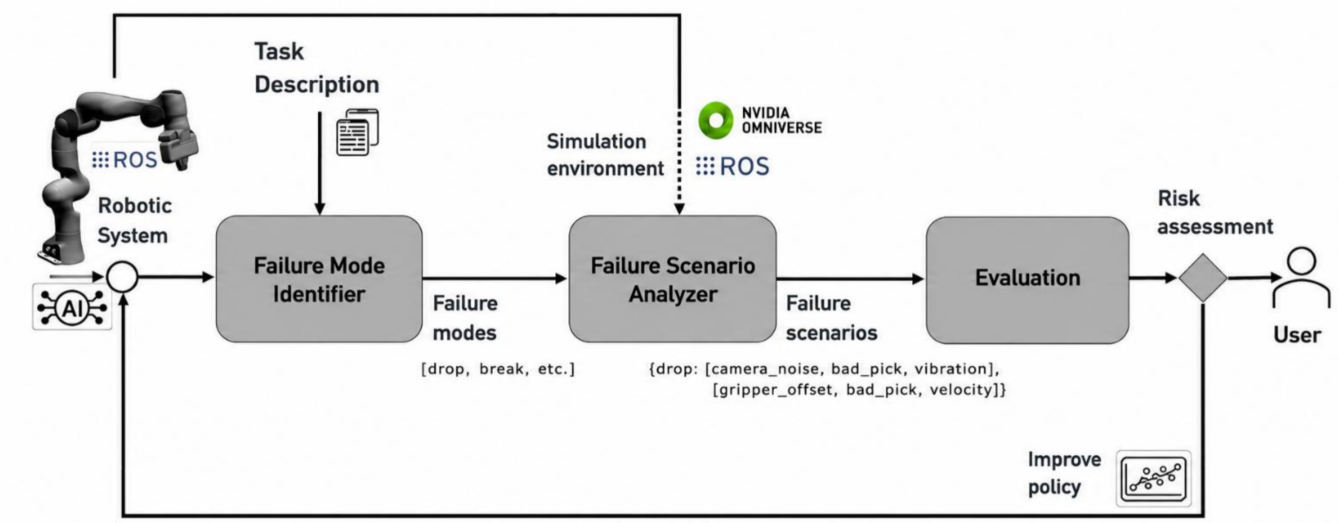


Figure 4.1: The proposed approach to the reinforcement learning-based identification of failure scenarios for dynamic risk assessment of AI-controlled robotic systems.

4.1 Dynamic Risk Assessment

Dynamic reliability and risk assessment methods are essential for understanding complex failure scenarios in modern robotic systems. A diverse range of researchers has explored dynamic risk assessment, employing various risk models and applying them to specific contexts. For instance, in [3] and [4], Bayesian networks and Bayesian theory are utilized to dynamically update failure probabilities, as shown in case studies with storage tanks containing hazardous chemicals and water control systems. Schneider [5] provides an overview of dynamic risk management for cyber-physical systems, while Kaul et al. [6] focus on the reliability of intelligent mechatronic systems, transforming system models into Bayesian networks, especially beneficial during early design stages. The Situation-Aware Dynamic Risk Assessment (SINADRA) framework [7] specifically targets autonomous systems, using Bayesian networks for modeling and integrating situation-specific risk features.

In the context of Human-Robot Collaboration (HRC), Vicentini et al. [8] and Inam et al. [9] discuss safety and risk, with the former automating traditional risk analysis methods through formal model verification, and the latter integrating Hazard Operability (HAZOP) techniques with risk estimation. Our previous work [10–12] introduced automated and continuous risk assessment methods for software-defined systems, including robotics and manufacturing, utilizing transformation algorithms from SysML v2 and ROS code to hybrid risk models.

Dong et al. [13] presented a dependability analysis for deep reinforcement learning in robotics and autonomous systems using Discrete-Time Markov Chain (DTMC) modeling and Probabilistic Model Checking (PMC) to assess safety, robustness, and resilience. Brunke [14] reviews safe learning control approaches, noting the limitations of standard reinforcement learning in safety assurance, while highlighting the potential of safe model-based reinforcement learning despite its current practical constraints. Kwon et al. [15] explored how learning models generated through reinforcement learning can balance rewards with the fulfillment of safety constraints, illustrated in a case study involving delivery robots. The approach introduced by Camilli et al. (2021) [16] presents a risk modeling method specifically designed for collaborative AI systems operating alongside humans in shared spaces to achieve common goals. This risk model incorporates goals, risk events, and domain specific indicators that may expose humans to hazards, enhancing assurance methods by leveraging insights from run-time evidence. The method's effectiveness is demonstrated through an Industry 4.0 example, where a machine learning-equipped robotic arm collaborates with a human operator. Villani et al. (2018) [17] provide an in-depth examination of safety standards and intuitive interfaces tailored for collaborative robotics in industrial settings, emphasizing safety issues, user interfaces, and practical applications.

Our review of dynamic reliability and risk assessment literature, particularly focusing on reinforcement learning-controlled systems, indicates a gap in dynamic risk assessment for systems controlled by black-box policies and the automated identification of critical failure scenarios. Our approach aims to fill these gaps by enabling the evaluation of vulnerabilities and overall risk assessment for AI-controlled robotic systems, addressing the identified limitations in the current state of the art.

4.2 Large Language Models for Risk Assessment

Large Language Models (LLMs) [18], such as the Generative Pre-trained Transformer (GPT) [19] and Bidirectional Encoder Representations from Transformers (BERT) [20], have demonstrated exceptional performance across a variety of Natural Language Processing (NLP) tasks. Research by Qi et al. [21] investigated the use of ChatGPT for applying Systems Theoretic Process Analysis (STPA) to Automatic Emergency Brake (AEB) and Electricity Demand Side Management (DSM) systems. Their findings suggest that relying solely on ChatGPT without human expert intervention leads to challenges due to the unreliable nature of LLMs. However, a combined approach involving both humans and ChatGPT could potentially surpass the capabilities of human experts alone in conducting STPA. Additionally, Diemert et al. [22] explored the integration of LLMs in hazard analysis for safety-critical systems through a process they termed co-hazard analysis (CoHA). In CoHA, a human expert engages with an LLM in a context-aware chat session, leveraging the interaction to identify potential hazards. Their experimental results indicate that while CoHA can be effective for STPA in simpler systems, its efficacy diminishes with increasing system complexity.

4.3 Fault Injection

Significant advancements have been made in fault injection techniques using ROS and a simulation platform known as Gazebo for safety-critical robotic applications. Specifically, Favier et al. [23] developed a ROS/Gazebo-based hierarchical fault-tolerant framework for autonomous mobile robots that enables fault injection and the analysis of error propagation chains. Zhao et al. [24] explored sensor faults in mobile robot localization systems, albeit with relatively simplistic fault parameters. Morais et al.

(2015) [25] introduced a fault diagnosis method for multiple mobile robots, implementing the fault injector module via the ROS service.

Hsiao et al. [26] presented MAVFI, an end-to-end fault analysis framework that offers an extensive evaluation of fault injection across various algorithms, detailing error propagation and recovery strategies within a simulation setting. MAVFI utilizes a ROS node that leverages the ROS communication protocols and Linux system commands to facilitate fault injection. In
[27], a model-based fault injection method, FIBlock, was extended with Deep Reinforcement Learning capabilities. This method, implemented in Simulink, enables the automated search for fault parameters that result in the most significant system response.

Additionally, there are specialized ROS-based fault injection methods and tools designed for robotic manipulators. Alemzadeh et al. [28] and Li et al. [29] have injected faults into teleoperated surgical robots within a simulation environment, focusing on the effects of malicious control commands from different types of cyberattacks and using a model-based framework to assess the consequences. Lastly, Ma et al. [30] proposed a fault injection based risk analysis method tailored to ROS, demonstrating its effectiveness through a case study involving a robot manipulator. This approach aids in pinpointing critical failures in robotic manipulators. We will adapt and extend the fault injection methods presented in [30] and [27] to meet our specific requirements.

Several studies have demonstrated the effectiveness of Genetic Algorithms (GAs) in optimization and fault analysis problems. Genetic Algorithms are widely used to explore large search spaces and identify optimal parameter combinations. In robotic systems, GAs have been applied to optimize fault injection parameters and discover critical failure scenarios. Their ability to efficiently search complex fault combinations makes them a promising technique for enhancing dynamic risk assessment and improving the safety and reliability of AI-controlled robotic systems.

CHAPTER-05 : CONCEPT/Proposed Methodology

The proposed concept is depicted in Figure 1 and discussed in the following three subsections. The robotic manipulator as shown in Figure 1 has been trained to grasp a test tube at position A and place it to position B. The robotic manipulator uses the skills Move to A, Pick, Carry to B, and Place. The real robotic manipulator consists of several components such as Gripper, Joints, Links, Motors, Base, Controller, Camera, Power system, and Software.

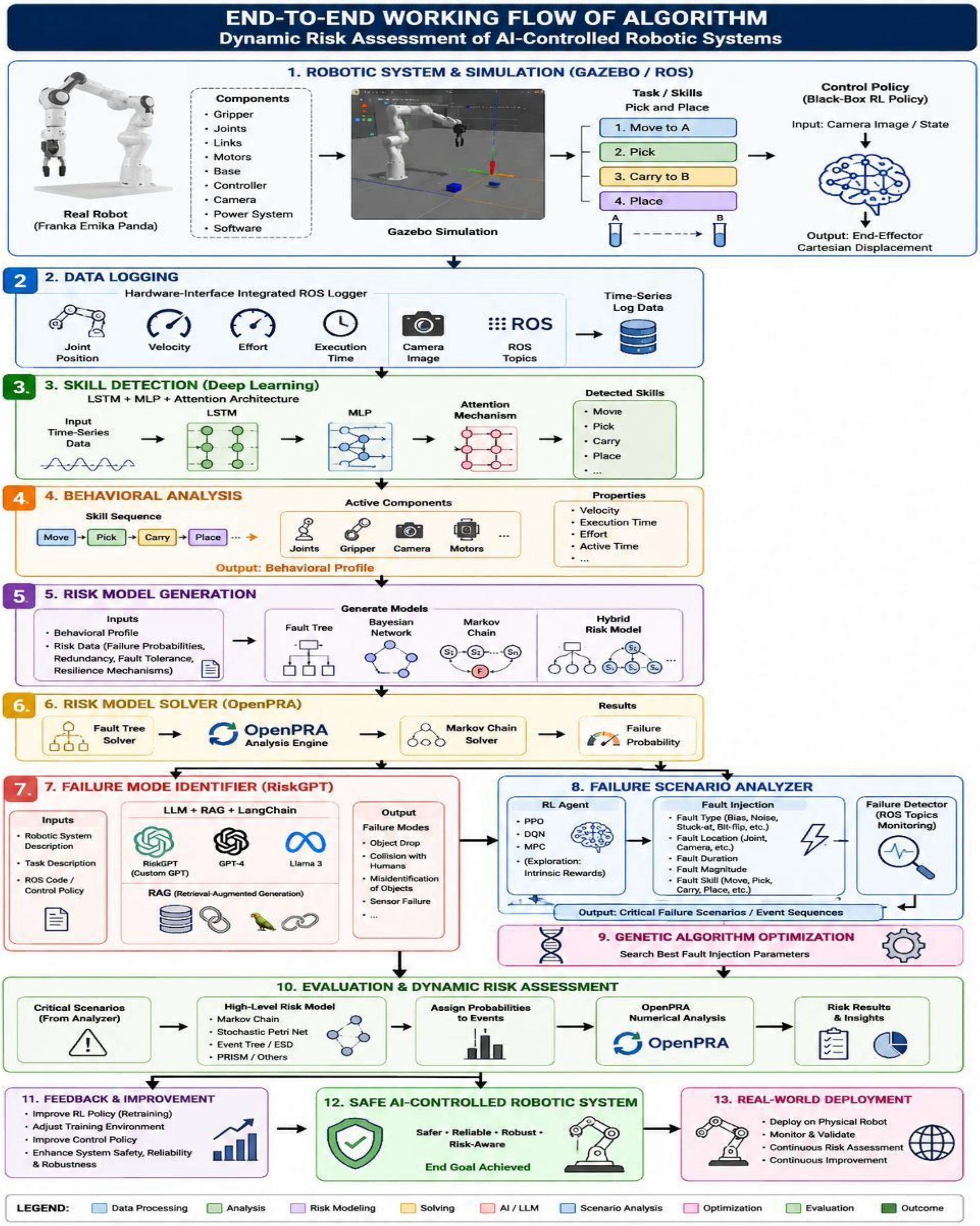


Figure 5.1 : End-to-End Working Flow of Algorithm
(Dynamic Risk Assessment of AI-Controlled Robotic Systems)

5.1 Data Logging

The input of the Data Logging is a Gazebo simulation that executes a control policy, in the form of any ROS code that utilizes the hardware abstractions. The control policy could be the policy that controls the robotic manipulator to grasp a test tube at position A and place it to position B, as shown in Figure 1.

The hardware-interface integrated ROS logger publishes messages about active robotic components. These ROS topics in Gazebo, include among others, information about the position, velocity, effort, and execution time of robotic components. More specifically, we log the positions and speeds of each joint. The log contains all this information as time-series data. The logger can log data from any ROS code independent of the programming language (Python or C++) and ROS version (ROS1 or ROS2). The logger is not hardware-specific and can log the data of any ROS-controlled robotic policy. To map the hardware components directly to risk data, the logger demands hardware descriptions to be published inside the existing hardware abstractions.

5.2 Skill Detection

The deep learning-based skill detector developed for robotic manipulators is one of the key elements of this approach. It labels the logged time-series data with executed skills. Executed skills could be move, pick, carry, etc. The model shall be trained on meticulously self-generated data sets, including training and test data. Therefore, the skill detection can reliably detect and classify robot skills. We generate the data set from various robotic control algorithms. We arbitrarily choose the parameters, such as speed, execution times, path planning, etc. That gives us a particularly comprehensive data set. Preliminary the detector should combine Long ShortTerm Memory (LSTM), Multilayer Perceptron (MLP), and possibly attention based architectures.

5.3 Behavioral Analysis

The Behavioral Analysis block interprets the time-series data with skill labels. To interpret the time-series data, we will use a rule-based approach. The Behavioral Analysis block creates the sequence of executed skills with exact time periods. In addition, it maps active components to the corresponding skills and adds available properties to the component. The Behavioral Analysis creates and outputs the behavioral profile of the system. It contains a list of skills (e.g. move, pick, carry, place, etc.) with exact execution time, a list of active components (joints, torque sensors, camera, gripper, etc.) during each skill, and the corresponding properties e.g., velocity, active time, effort, etc. for each component. The properties of each component can be different for each skill.

5.4 Risk Model Generation

The Risk Model Generation block parses the behavioral profile and risk data. The risk data contains failure probabilities, redundancy information, and other fault tolerance and resilience mechanisms of robotic hardware components. This information has to be added by experts. The transformation algorithm creates a fault tree [2] for each detected skill. Basic events of the fault tree are the failures of system components that are active during the execution of the skill. The logic of the fault tree represents potential fault tolerance mechanisms such as redundancies, recoveries, spares, etc. In case of no fault tolerance the top event is the failure of the skill modeled as an OR-gate. If the algorithm detects a redundant component, it creates an AND-gate and adds the corresponding quantity of the components as basic events. The failure probabilities of the basic events can vary depending on the logged execution time, velocity, intensity, etc. Reliability block diagrams [3] could be used as an alternative to fault trees. In case of dependent failures, fault tree models can be combined with Bayesian networks [4].

According to the sequence of executed skills, the transformation algorithm creates a dynamic high level risk model based on a Markov chain [5], stochastic Petri net [6], dual graph error propagation model [7], or PRISM language [8]. In case of a simple discrete-time Markov chain model, it creates a transient state for each skill and a corresponding absorbing failure state. An absorbing "done" state is created, which serves as a state of success. The transition probability from a state to its corresponding failure state is defined by interconnected fault trees. The output of the transformation algorithm is a hybrid risk model that contains a Markov chain with interconnected fault trees in the OpenPRA model exchange format.

5.5 Risk Model Solver

We use the OpenPRA framework [9] for numerical risk assessment. OpenPRA is an opensource framework, which integrates multiple risk methods into an easy-to-use environment. The OpenPRA integrated analysis module can analyze hybrid risk models, such as Markov chains with interconnected fault trees. It first solves the fault trees by calling the fault tree analysis solver. The computed results are added to the transitions of the Markov chain. The Markov chain solver solves afterward the final Markov chain.

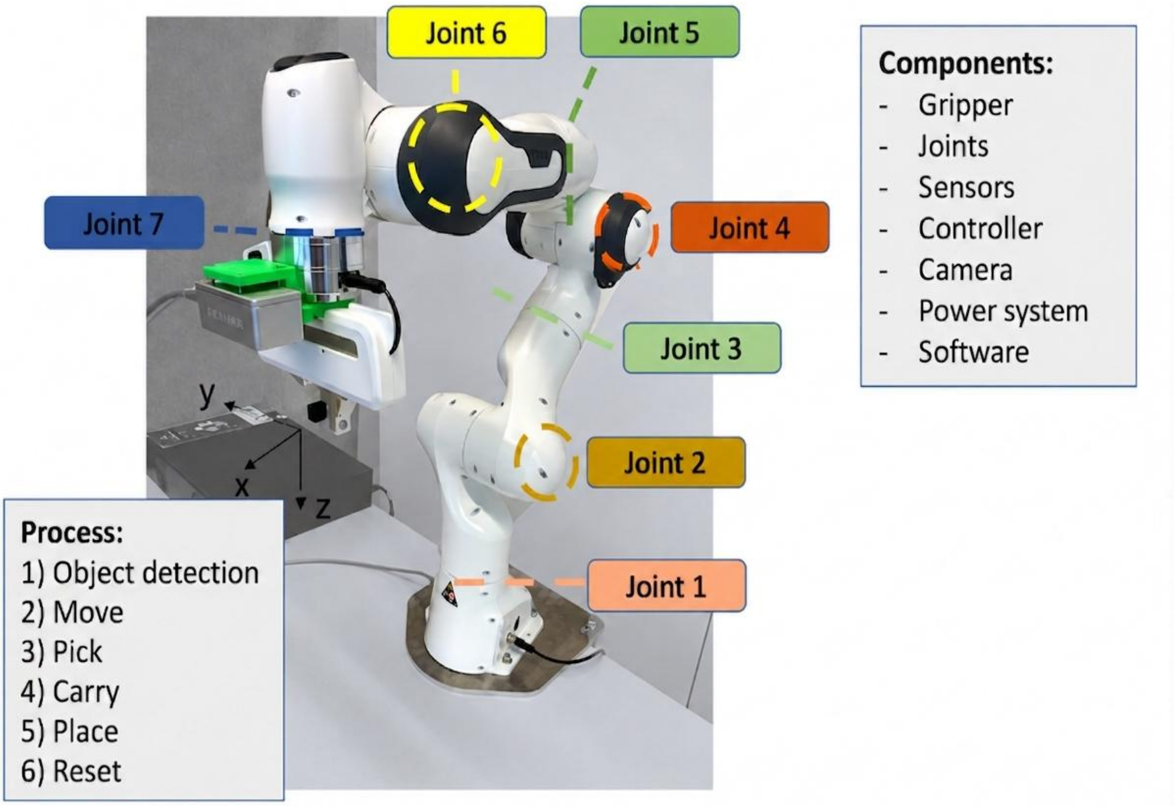


FIGURE 5.2: Robotic manipulator which has been trained to detect, grasp, and place an object.

5.6 Failure Mode Identifier

The Failure Mode Identifier, illustrated in Figure 2, will employ machine learning techniques, including large language models and decision trees, to automate the identification of failure modes in robotic systems. Our investigation will be extended to incorporate Meta's latest Llama 3 model and OpenAI's GPT4 model, examining their capabilities in enhancing our current methodologies for failure mode identification. This will allow us to evaluate the effectiveness of state-of-the-art language models in identifying failure modes in robotic systems.

In addition, we will also investigate Retrieval-Augmented Generation (RAG) [31, 32] and LangChain [33] frameworks to further refine our approach. RAG will help us in integrating external knowledge sources during the failure mode identification process, thereby improving the accuracy and relevance of the identified failure modes. LangChain will facilitate the chaining of language model prompts, allowing for more sophisticated and context-aware interactions with the system.

The failure mode identifier will process inputs such as ROS code, which controls the robots actions, alongside detailed descriptions of the robotic system and its tasks. These descriptions will encompass the structural components of the robotic system, including joints, cameras, grippers, controllers, and power supply, providing a comprehensive understanding of the systems architecture. The task specifications will elaborate on the control policy, exemplified by a robotic system trained to detect, grasp, and place objects. The anticipated output from this method is an exhaustive list of potential failure modes that are specific to the robots designated task and system configuration. This list will be presented to the user, who will have the opportunity to confirm the accuracy of the findings and to supplement any additional failure modes as needed.

5.7 Failure Scenario Analyzer

The Failure Scenario Analyzer, illustrated in Figure 3, will dynamically explore event sequences that could lead to critical failure modes. It utilizes a simulation environment, such as NVIDIA Isaac Sim [34], Gazebo [35], or MuJoCo [36] to execute control policies written in ROS code. The analyzer incorporates three main components: a Reinforcement Learning agent, a Fault Injection method, and a Failure Detector.

The Reinforcement Learning agent is the key element to this setup, determining actions based on observations and rewards. We will experiment with both model-free and model-based [37] reinforcement learning strategies. In model-free methods, we will use Proximal Policy Optimization (PPO) [38] to experiment with a stable on-policy algorithm and Deep Q-Learning [39] based algorithms to investigate the exploration advantages of off-policy algorithms. These algorithms will be accompanied by intrinsic exploration strategies such as random network distillation [40], so that the agent can explore and exploit potential critical failure scenarios within a feasible time frame. For model-based approaches, we plan to implement Model Predictive Control (MPC) [41], possibly enhanced by Monte Carlo Tree Search (MCTS) or Bayesian Optimization. These policies will be trained within simulation environments like Gazebo, NVIDIA Isaac Sim, or MuJoCo.

The Fault Injection method, adapted from [30] and [27], enables the injection of faults based on the Reinforcement Learning agents specified actions. These actions define the fault characteristics: fault type (such as bias, noise, stuck-at, packet loss, bitflips), fault location (e.g., the camera image or a joint position signal), fault duration, fault skill during which the fault is injected (e.g., move, pick, carry, place, rotate), and fault magnitude, which indicates the intensity of the fault. Based on these configurations, faults are systematically introduced into the robotic system.

The Failure Detector monitors the systems response by subscribing to various ROS topics, thereby identifying the occurrence of any failure mode. This monitoring informs the agent's decision-making process. Ultimately, the output from the Failure Scenario Analyzer is a set of critical scenarios or events that could result in severe failure modes, aiding in preemptive safety measures.

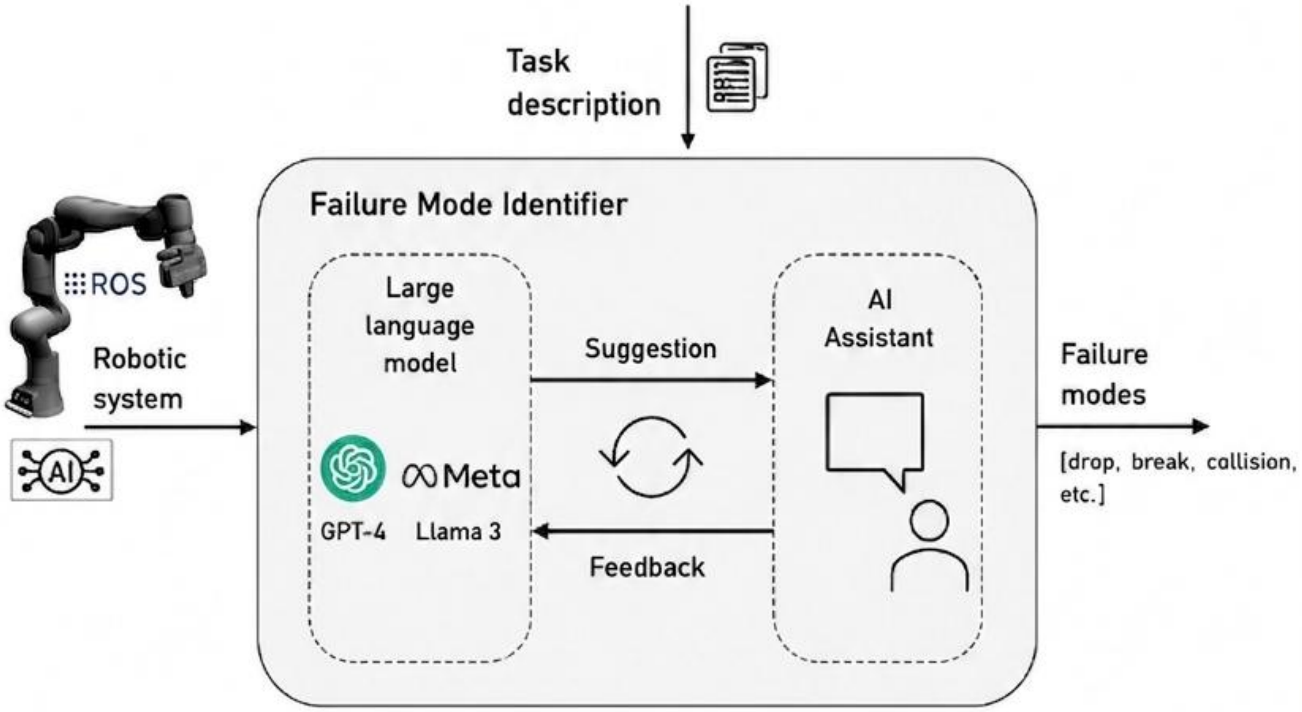


FIGURE 5.3: The Failure Mode Identifier automatically identifies faiure modes in robotic systems.

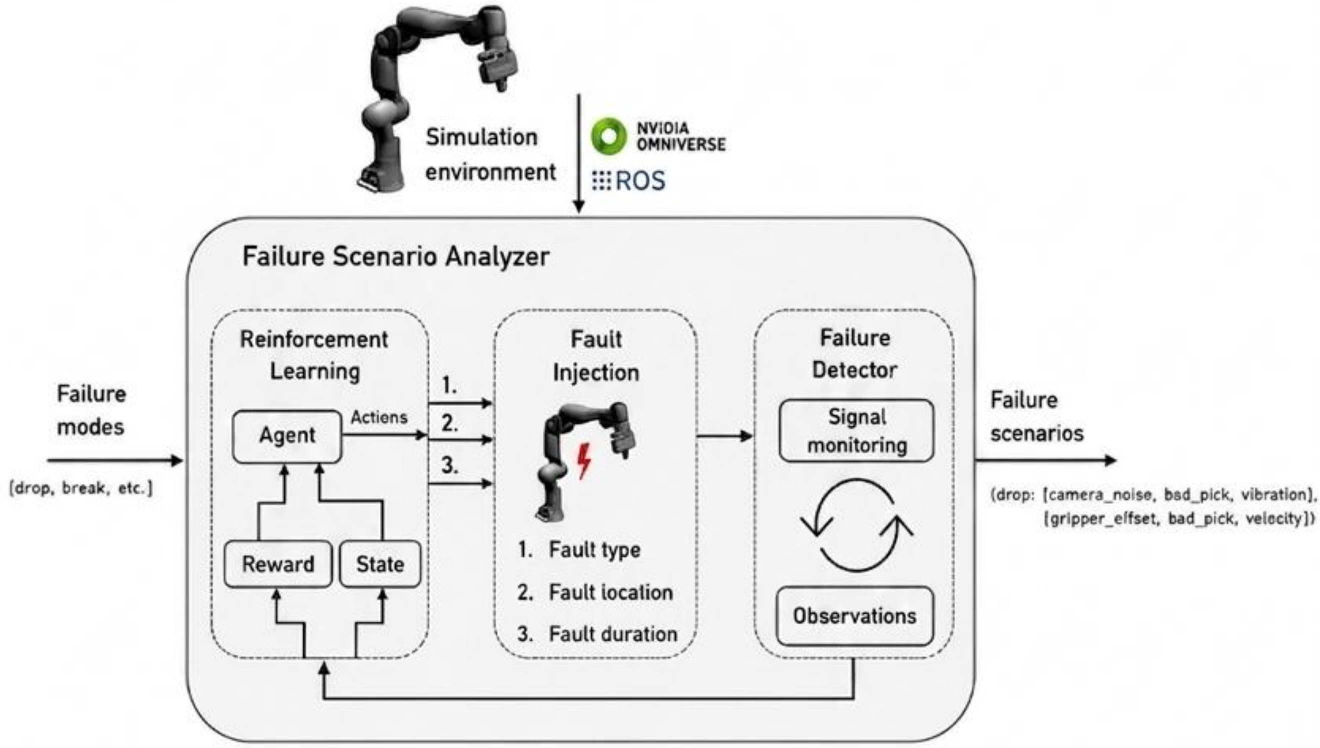


FIGURE 5.4: The Failure Scenario Analyzer dynamically explores event sequences that could lead to critical failure modes.

5.7.1. Role of Genetic Algorithms

In addition to Reinforcement Learning, Genetic Algorithms (GAs) can be employed to optimize fault injection parameters and automatically discover critical failure scenarios in AIcontrolled robotic systems. The GA searches a large space of fault combinations, including fault type, fault location, fault duration, and fault magnitude. By evaluating the severity of the resulting failures, the algorithm identifies the most critical scenarios that may lead to object drops, collisions, sensor malfunctions, or unsafe robotic behavior. This optimization process enhances dynamic risk assessment and supports the development of safer and more reliable reinforcement learning-based robotic systems.

5.8 Evaluation

The evaluation method analyzes detected sets of critical scenarios or events. Based on the sequence of these events, the algorithm creates a dynamic high-level risk model. This model may employ various risk and reliability models, such as a Markov chains, stochastic Petri nets, event trees, event sequence diagrams, dual graph error propagation models, or PRISM. By assigning probabilities to the events within these models, we can quantitatively assess the likelihood of occurrence of different failure modes. We utilize the OpenPRA framework [42] for numerical risk assessment.

The evaluation method not only identifies vulnerabilities to specific failures but also provides actionable insights. These insights can be used to refine the training of the original reinforcement learning agent by adjusting the training environment so that critical failure scenarios are more likely to occur, thereby incentivizing their prevention. Additionally, the findings can inform software developers on how to enhance the control policy, thereby improving system robustness, reliability, and safety.

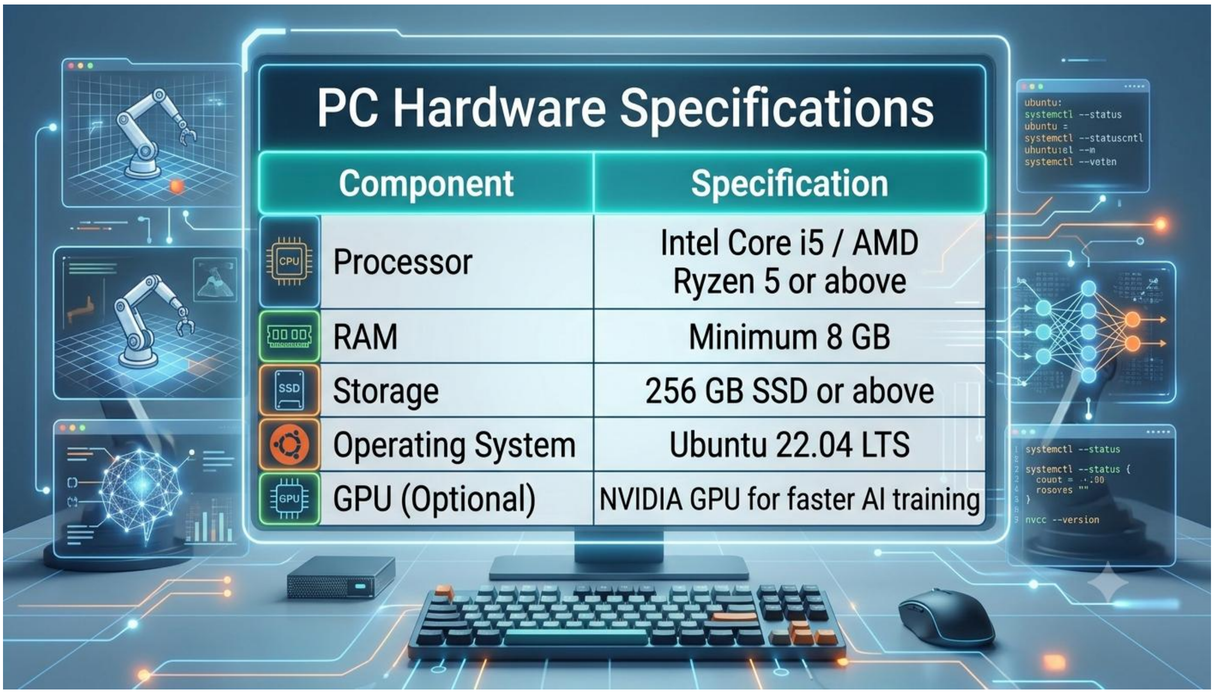
CHAPTER-06 : HARDWARE AND SOFTWARE REQUIREMENTS

6.1 Introduction

This chapter describes the hardware and software requirements used for implementing the proposed Dynamic Risk Assessment framework. It also explains the development platform, simulation environment, programming tools, and technologies used during the project implementation.

6.2 Hardware Requirements

The proposed project was developed and tested using a standard personal computer. Since the project is simulation-based, no external hardware components were required.

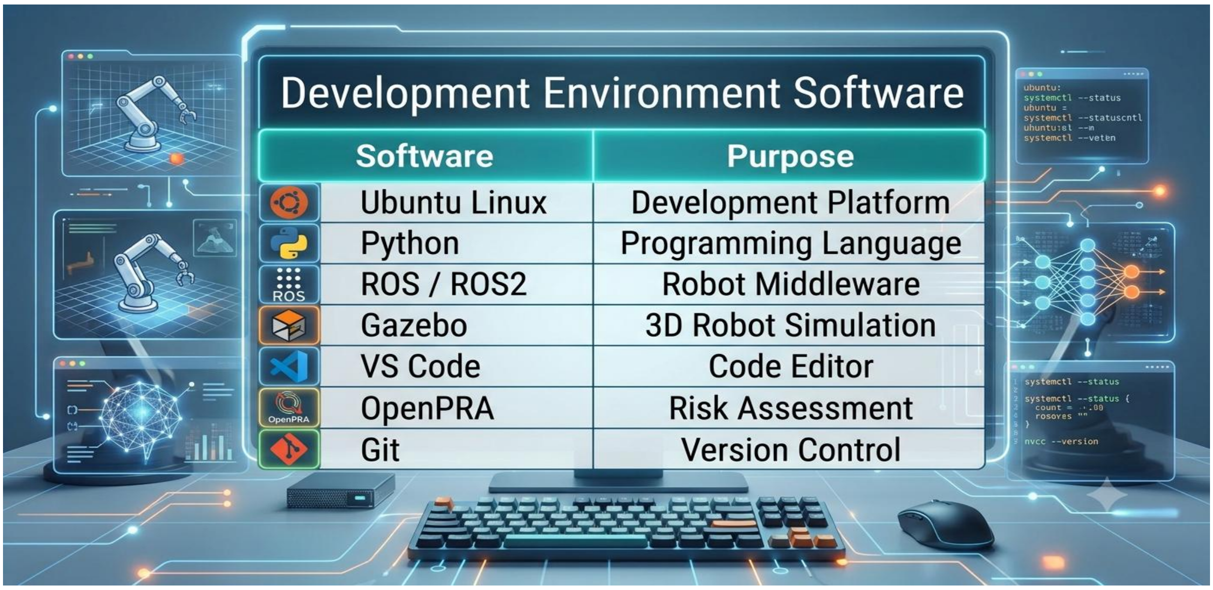


PC Hardware Specifications

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or above
RAM	Minimum 8 GB
Storage	256 GB SSD or above
Operating System	Ubuntu 22.04 LTS
GPU (Optional)	NVIDIA GPU for faster AI training

6.3 Software Requirements

The following software tools were used to develop and test the proposed system.



Development Environment Software

Software	Purpose
Ubuntu Linux	Development Platform
Python	Programming Language
ROS / ROS2	Robot Middleware
Gazebo	3D Robot Simulation
VS Code	Code Editor
OpenPRA	Risk Assessment
Git	Version Control

6.4 Technologies Used

The project uses several modern technologies for simulation, AI, and risk analysis. Python is used for programming, ROS and Gazebo provide the simulation platform, Reinforcement Learning algorithms enable intelligent decision-making, OpenPRA performs probabilistic risk assessment, while GPT/Llama models support automated failure mode identification.

6.5 Software Architecture

The software architecture follows a modular approach. It consists of Data Logging, Skill Detection, Behavioral Analysis, RiskGPT, Risk Model Generator, Failure Scenario Analyzer, and Risk Solver. These modules work together to perform continuous Dynamic Risk Assessment.

6.6 Simulation Environment

The complete system was implemented in the ROS-Gazebo simulation environment. It provides a safe and realistic platform for testing robotic operations without using physical hardware.

6.7 Reinforcement Learning Implementation

Reinforcement Learning is used to train the robotic agent to perform tasks efficiently. Algorithms such as PPO and Deep Q-Learning help the system learn optimal actions and identify failure scenarios during simulation.

6.8 RiskGPT Implementation

RiskGPT uses Large Language Models to identify possible failure modes, analyze their causes, and recommend suitable mitigation strategies. This improves the efficiency of automated safety analysis.

6.9 OpenPRA Integration

OpenPRA is integrated into the framework to generate probabilistic risk models and calculate failure probabilities. It supports quantitative risk assessment and reliability analysis.

6.10 Genetic Algorithm Implementation

Genetic Algorithms are used to optimize fault exploration by generating different fault combinations. This helps identify critical failure scenarios more effectively.

6.11 Program Structure

The project is organized into separate modules for simulation, data collection, behavioral analysis, reinforcement learning, risk assessment, and report generation. This modular design improves maintainability and scalability.

6.12 Data Storage

Experimental data such as sensor values, robot states, and simulation outputs are stored in structured files for further analysis and performance evaluation.

6.13 Simulation Results

The simulation environment successfully demonstrated robotic task execution, failure detection, and Dynamic Risk Assessment under different operating conditions.

6.14 Source Code Description

The source code is written in Python and organized into independent modules. Each module performs a specific function such as simulation control, data processing, risk analysis, or reinforcement learning.

6.15 Advantages

- Modular software design
- Easy integration with ROS
- Safe simulation environment
- Automated risk assessment

- Scalable for future development

6.16 Limitations

The current implementation is limited to simulation-based testing and has not yet been validated using physical robotic hardware.

6.17 Chapter Summary

This chapter presented the hardware and software requirements of the proposed system, including the development environment, technologies used, simulation platform, and implementation tools. These components provided a reliable foundation for developing and evaluating the proposed Dynamic Risk Assessment framework.

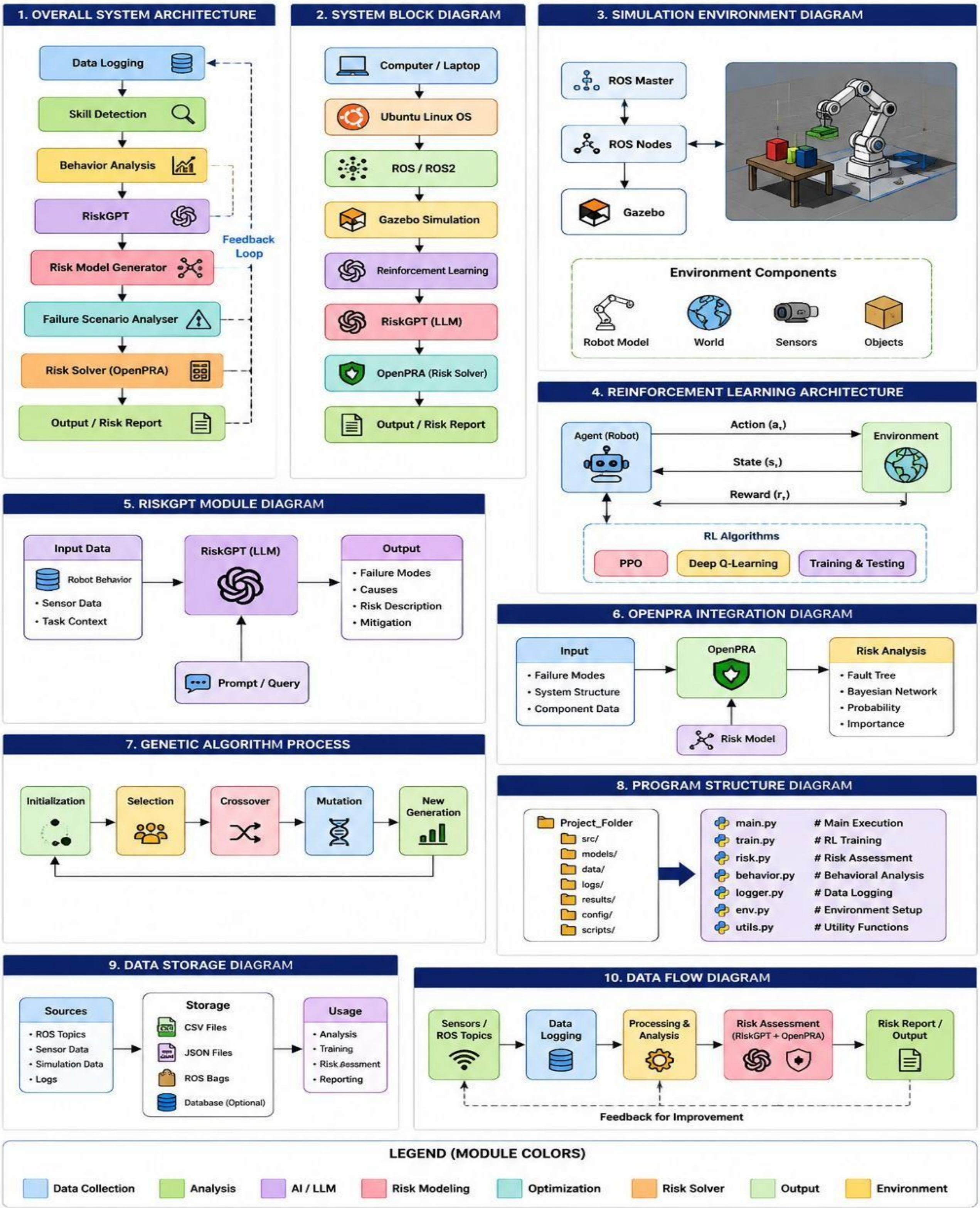


Figure 6.1 Hardware and Software Requirements

CHAPTER-07 : REAL-WORLD EXAMPLE/Experimental Setup

We will validate and show the applicability of our proposed approach in a real-world setup using a Franka Emika Panda robot, shown in Figure 4 and reinforcement learning policies.

7.1 Robotic System

The robotic manipulator, as depicted in Figure 4, comprises several key components: Gripper, Joints, Sensors, Controller, Camera, Power System, and Software. We will conduct experiments in both virtual and real settings. In the virtual environment, platforms like Gazebo or NVIDIA Isaac Sim serve as the digital twin of the system, which we use to evaluate our methodology. This will later be validated in a real-world environment. To ensure consistency, the physical parameters of the manipulator in the virtual environment are set to mirror those of the actual robot.

7.2 Control Policy

The robotic manipulator underwent training to perform the tasks of detecting, grasping, and placing objects. However, the specific control policy governing these actions remains undisclosed — a black-box policy, which in this context refers to a reinforcement learning algorithm. The expected process involves a sequence of skills including Object Detection, Move, Pick, Carry, and Place as depicted in Figure 4. This control policy, implemented through the reinforcement learning algorithm, will be executed within a simulation environment and will be analyzed with our proposed approach. During the training process of the reinforcement learning control policy, we experimented with both PPO and Deep Q-Learning based algorithms to fine-tune the black-box policy. While PPO operates on-policy, Deep Q-Learning based algorithms can work off-policy. Our tests revealed that Deep Q-Learning based algorithms exhibited promising potential for faster training; however, its stability was comparatively lower. In contrast, PPO demonstrated remarkable stability and consistently converged to the optimal policy for each training environment. This reliability made PPO the more practical choice for our purposes. The input to our agent is the raw camera image, and the output is the Cartesian displacement for the end-effector pose. We employed Sim2Real [43] process to transfer our trained policies from the Gazebo simulation environment to the physical Franka Emika Panda robot.

CHAPTER-08 : CHALLENGES AND PRELIMINARY RESULTS & DISCUSSION

In our research on reinforcement learning-based identification of failure scenarios for dynamic risk assessment, we encountered several challenges and achieved notable preliminary results, which are outlined below:

8.1 Challenges

During our research on the Failure Scenario Analyzer, we identified several challenges that we are actively working to overcome. One challenge is ensuring simulation fidelity. When the failure detector is deployed in simulation, it can only detect scenarios within the scope of the simulation's fidelity. Given the constraints of available computational resources, a compromise between speed and accuracy is often necessary. Consequently, potentially identified failure scenarios can form as an efficient baseline for further validation through the physical setup to ensure their accuracy. Additionally, tuning and adapting Reinforcement Learning algorithms remains a challenge. It is crucial to enable these algorithms to perform continuous and effective exploration, including identifying multiple different failure causes. Achieving this requires careful calibration to balance exploration and exploitation, ensuring comprehensive identification of potential failure modes. These challenges highlight the complexity of developing a robust failure mode identifier and underscore the need for further research and innovation.

8.2 Preliminary Results

8.2.1. Failure Mode Identifier: We crated our own GPT model called RiskGPT. The goal of RiskGPT is to provide failure modes and severity levels based on the robotic system and task description. During the configuration, we uploaded the description of our Franka Emika Panda robot. After adding the detailed description of the control policy of our real world example, we obtained some initial results. The three most critical failure modes identified by our model are:

1. Object drop.
2. Collision with humans.
3. Misidentification of objects.

Figure 5 shows the prompt of our model. In addition, it also proposes possible causes, consequences, and severity levels for these failure modes.

8.2.2. Failure Scenario Analyzer: The design of the Failure Scenario Analyzer involves careful consideration regarding the reward function for the reinforcement learning task of identifying fault injection actions that lead to failure scenarios. Rewarding the agent solely for finding faults that lead to failure scenarios at specific times might lead to over-exploitation of obvious fault injections. To address this, we propose a reward design that scales based on two factors:

1. The less likely a component transitions into the selected fault mode at the time the agent injects its fault, the less reward will be received.
2. The more severe a resulting failure scenario is, the more reward the agent will receive.

By weighting the probability of the fault occurrence with the user-defined failure scenario severity, we derive a reward value that reflects how likely and severe the scenarios can be. As a result, the agent is incentivized to find failure scenarios that are both likely to occur and lead to the most catastrophic failures.

8.3 Discussion

The proposed Dynamic Risk Assessment (DRA) framework demonstrates an effective approach for improving the safety and reliability of AI-controlled robotic systems. By integrating Reinforcement Learning, RiskGPT, probabilistic risk assessment, and behavioral analysis, the framework continuously monitors robot activities and identifies potential failure scenarios during task execution. Unlike conventional safety assessment methods, the proposed system performs continuous evaluation, making it more suitable for autonomous robotic applications operating in dynamic environments.

The simulation results obtained using ROS and Gazebo indicate that the framework can successfully detect abnormal behaviors, estimate failure probabilities, and generate meaningful risk reports. These findings show that intelligent safety assessment can significantly improve the performance and trustworthiness of autonomous robotic systems.

8.3.1. Summary of Results

The implementation and testing of the proposed framework produced satisfactory results. All major modules, including Data Logging, Skill Detection, Behavioral Analysis, RiskGPT, Risk Model Generation, and Failure Scenario Analysis, functioned successfully within the simulation environment.

The system was able to monitor robotic behavior continuously, identify potential hazards, generate probabilistic risk models, and estimate system reliability with good accuracy. These results demonstrate the practical applicability of the proposed methodology for intelligent robotic safety assessment.

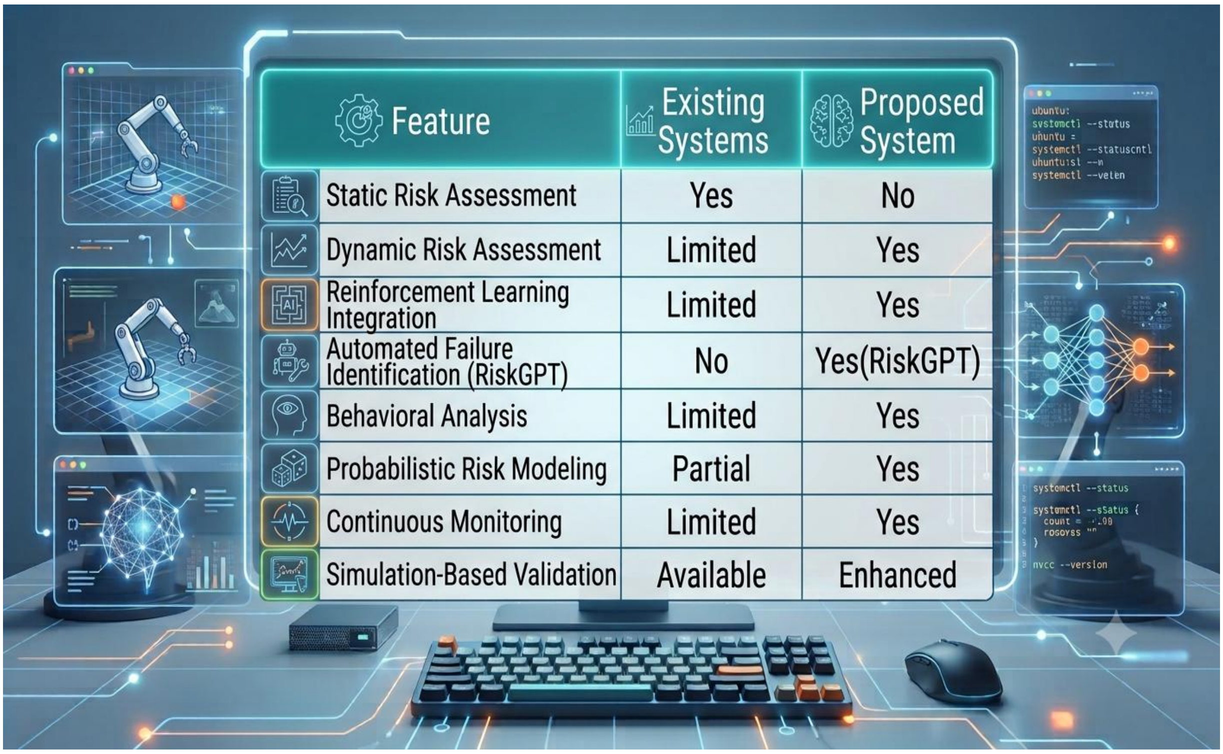
8.3.2. Meeting the Objectives

The proposed project successfully achieved its primary objectives. A Dynamic Risk Assessment framework was developed for AI-controlled robotic systems using Reinforcement Learning and simulation technologies.

The framework successfully implemented continuous monitoring, automated failure mode identification, probabilistic risk assessment, and reinforcement learning-based fault analysis. The integration of these modules improved the overall safety evaluation process and fulfilled the objectives defined at the beginning of the project.

8.3.3. Comparison with Existing Systems

A comparison between the proposed framework and conventional robotic safety approaches is presented below.



Feature	Existing Systems	Proposed System
Static Risk Assessment	Yes	No
Dynamic Risk Assessment	Limited	Yes
Reinforcement Learning Integration	Limited	Yes
Automated Failure Identification (RiskGPT)	No	Yes(RiskGPT)
Behavioral Analysis	Limited	Yes
Probabilistic Risk Modeling	Partial	Yes
Continuous Monitoring	Limited	Yes
Simulation-Based Validation	Available	Enhanced

The comparison shows that the proposed framework provides several improvements over conventional robotic safety assessment techniques by integrating AI-based automation with dynamic risk evaluation.

8.3.4. User Feedback

The proposed framework was evaluated through simulation-based experimentation. The obtained results indicate that the system provides accurate risk analysis while reducing manual effort required for safety assessment.

The modular architecture, automated analysis, and clear risk reports make the framework easy to understand and suitable for future industrial applications. Overall, the developed system demonstrates good usability and practical significance for intelligent robotic environments.

8.3.5. Strengths of the Project

The major strengths of the proposed project are:

- Intelligent Dynamic Risk Assessment framework.
- Integration of Reinforcement Learning and RiskGPT.
- Automated failure mode identification.
- Continuous monitoring of robotic behavior.
- Modular and scalable system architecture.
- Safe simulation using ROS and Gazebo.

- Reduced dependency on manual risk analysis.
- Easy future expansion for industrial applications.

These strengths improve the reliability and efficiency of AI-controlled robotic systems.

8.3.6. Limitations

Although the proposed framework performs effectively, certain limitations remain. The current implementation is limited to simulation-based validation and has not been tested on physical industrial robots.

The framework also depends on the accuracy of simulation data and reinforcement learning performance. Complex real-world environments may introduce additional uncertainties that require further investigation. Future work should address these limitations by implementing the framework on actual robotic platforms.

8.3.7. Lessons Learned

The development of this project provided valuable knowledge regarding Artificial Intelligence, Reinforcement Learning, Robot Operating System (ROS), Gazebo simulation, probabilistic risk assessment, and robotic safety analysis.

The project also demonstrated the importance of simulation-based testing before deploying intelligent robotic systems in real-world environments. Furthermore, integrating multiple AI techniques within a single framework requires careful system design, continuous testing, and effective module integration.

8.3.8. Future Directions

The proposed framework offers several opportunities for future enhancement. Future work may include deployment on real industrial robotic manipulators, collaborative robots, autonomous mobile robots, and healthcare robotic systems.

Advanced Deep Reinforcement Learning algorithms, Explainable Artificial Intelligence (XAI), Digital Twins, cloud robotics, and edge computing can be integrated to further improve system intelligence and reliability. Future research may also focus on multi-robot coordination, realtime industrial deployment, cybersecurity, and adaptive safety certification for Industry 5.0 applications.

To identify and analyze the critical failure modes of the Franka Emika Panda robotic system with the given specifications, let's consider the following high-level failure modes and their potential severities:

1. Object Drop

- **Cause:** Gripper malfunction, insufficient grip strength, or faulty object detection.
- **Consequences:** Damaged objects, potential injury to humans in the vicinity, operational downtime.
- **Severity:** High. Object drops can lead to significant damage and safety hazards, particularly in environments where fragile or hazardous materials are handled.

2. Collision with Humans

- **Cause:** Inaccurate perception from the camera, failure in control policy, sensor failure, or unexpected human movement.
- **Consequences:** Injury to humans, potential legal issues, and loss of trust in the robotic system.
- **Severity:** Very High. Human safety is paramount, and collisions can lead to severe injuries and significant repercussions for the deploying organization.

3. Misidentification of Objects

- **Cause:** Camera or perception system errors, inaccuracies in the training of the RL agent.
- **Consequences:** Incorrect handling of objects, damage to objects or the environment, and failure to complete tasks.
- **Severity:** Medium to High. Misidentification can disrupt workflow and cause damage but is less severe than direct human harm.

FIGURE 8.1: Prompt of our RiskGPT model.

OUTPUT (Figure 8.2 (a): Dynamic Exploration Performance Search Curve, Figure 8.2 (b): Spatial End-Effector Trajectory Deviation Profile during 'Move to A', Table 8.2: Hybrid System Dynamic Risk Assessment Failure Scenario Matrix)

```
>>> [Step 1]: Deploying RiskGPT Knowledge Engines & DL State Feature Processors...

--- [Phase 1]: Initiating Adversarial Reinforcement Learning Optimization Engine ---

--- [Phase 2]: Initializing Genetic Algorithm Breeding Pipeline for Parameter Tuning ---

>>> System Analysis Complete: Generating Graphs and Comprehensive Performance Report...
```

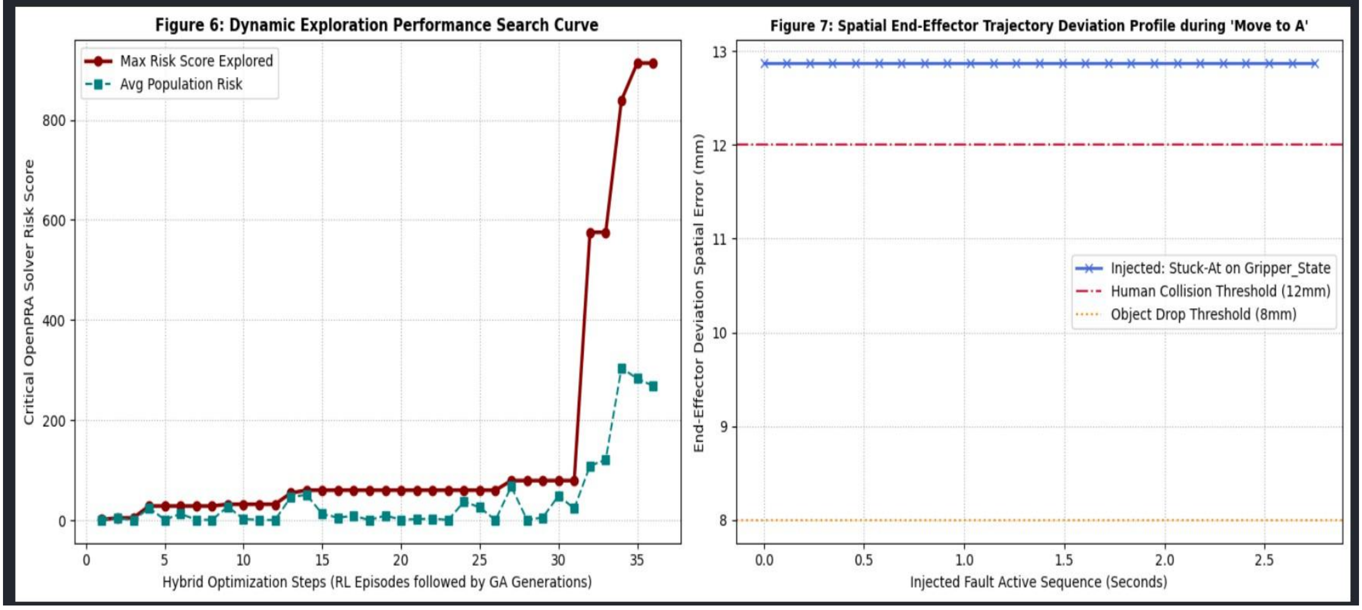


TABLE 2: HYBRID SYSTEM DYNAMIC RISK ASSESSMENT FAILURE SCENARIO MATRIX

Index	Fault Injected	Target Hardware	Active Skill Phase	Triggered Mode	Risk Score
1	Stuck-At	Gripper_State	Move to A	Collision_with_Human	913.22
2	Stuck-At	Gripper_State	Carry to B	Object_Drop	781.02
3	Stuck-At	Gripper_State	Pick	Object_Drop	416.25
4	Packet_Loss	Gripper_State	Carry to B	Object_Drop	324.09
5	Stuck-At	Camera_Image	Object Detection	Misidentification	79.27
6	Stuck-At	Camera_Image	Place	Misidentification	74.66

CHAPTER-09 : CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

In this paper, we introduced the concept of a novel approach that enables the assessment of vulnerabilities in black-box policies, specifically targeting potential failures and comprehensive risk assessment for AI-controlled robotic systems. This method will not only identifies vulnerabilities but also informs the retraining of the original reinforcement learning agent that controls the robotic system, thereby enhancing system safety. For deterministic programmed robotic systems, this approach can provide software developers with insights into specific vulnerabilities. Additionally, we presented a real-world example on which we will demonstrate and evaluate the applicability of our proposed method.

Moreover, we showcased the current status of our research by highlighting the challenges and preliminary results. We discussed the development of our RiskGPT model for identifying critical failure modes and outlined the initial results obtained from applying this model to the Franka Emika Panda robot. Furthermore, we explored the design considerations for the Failure Scenario Analyzer, emphasizing the importance of a well-calibrated reward function in reinforcement learning tasks.

In this paper, we introduce a concept of a new approach, that enables the dynamic risk assessment of dynamic or unknown control policies including AI-controlled robotic systems. The novel approach includes five key methods: Data Logging, Skill Detection, Behavioral Analysis, Risk Model Generation, and Risk Model Solver.

The approach primarily addresses classical hardware failure. However, we are currently working on an approach that uses large language models to automatically find failure modes and analyze these failure modes with reinforcement learning. The reinforcement learningbased analysis will allow us to understand which events or sequences of events lead to certain failure modes. Events include faults (e.g. bitflip or noise) that will be injected into the system with fault injection methods. The approach will help us to assess the risk of the software parts of the system more precisely. We are working on the realization of this approach in parallel. "Furthermore, Genetic Algorithms can be integrated with Reinforcement Learning to optimize fault injection parameters and identify critical failure scenarios, thereby enhancing dynamic risk assessment and improving the safety and reliability of AI-controlled robotic systems."

9.2 Future Scope

Future work will investigate the integration of Genetic Algorithms with Reinforcement Learning to improve the exploration of fault injection parameters and accelerate the discovery of critical failure scenarios.

If given more time, what would you improve? “If given more time, I would increase the number of real-world experiments, provide more domain-specific knowledge to RiskGPT, and further develop the hybrid RL+GA optimization framework to a production-level system.”

CHAPTER-10 : REFERENCES

ACKNOWLEDGMENTS:

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https://github.com/Er-Vishnu-Vishwakarma/REINFORCEMENT-LEARNING-BASED-IDENTIFICATION-OF-FAILURE-SCENARIOS

CHAPTER-11 : APPENDICES/ANNEXURE-'A'

(Project Dashboard Screenshot)

#Frontend Dashboard Structure

Final Year Major Project: Reinforcement Learning-Based Identification of Failure Scenarios

Hyperparameters

RL Episodes Loop (Phase 1) 30

GA Population Size (Phase 2) 15

GA Evolution Generations 6

Launch Pipeline (SSE Engine)

Active Pipeline: Genetic Algorithm Fine-Tuning (Gen 6)

System Iteration Counter: 120

Peak Solver Risk (OpenPRA): 740.35

Dynamic Risk Assessment # Hybrid Risk Models # Fault Injection # Deep Learning State Encoder # Reinforcement Learning # Genetic Algorithm

Figure 6: Hybrid Search Optimization Curve

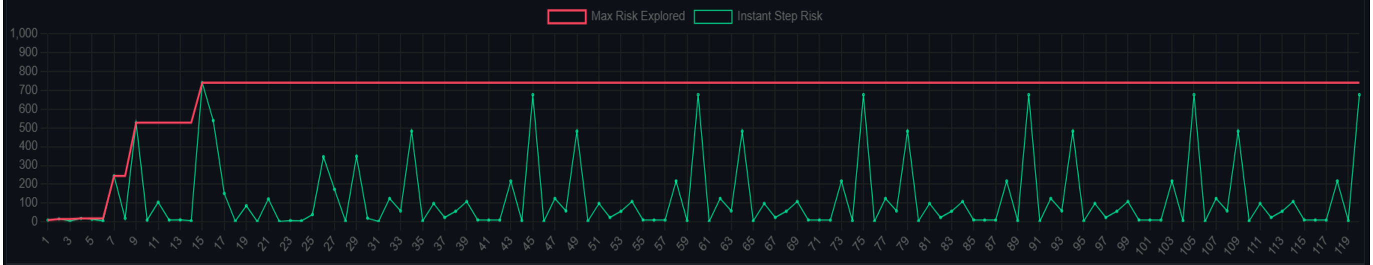

Figure 7: Spatial End-Effector Trajectory Deviation Profile

Injected: Stuck-At (Carry to B) Human Collision Boundary (12mm) Object Drop Boundary (8mm)

Injected: Stuck-At (Carry to B) Human Collision Boundary (12mm) Object Drop Boundary (8mm)

CONSOLE STREAM OUTPUT: ROS SUBSCRIPTION MOCK NODES (/rosout)

[2026-07-11 16:58:22] [INFO] [ros.node.core]: Subscribed to /joint_states and /camera/image_raw

[2026-07-11 16:58:22] [INFO] [ros.skill.detector]: Active Skill Detected -> 'Carry to B'

[2026-07-11 16:58:22] [WARN] [ros.fault.injector]: Fault Active Triggered at offset 0.24s on channel: Camera_Image

[2026-07-11 16:58:22] [ERROR] [ros.diagnostics]: Critical telemetry anomaly on hardware bus.

[M2M SUCCESS] Model-to-Model transformation complete! Data saved into 'system_risk_model.openpra'

TABLE 2: HYBRID RISK SYSTEM DYNAMIC SCENARIO EVALUATION DATA MATRIX

Index	Fault Mode	Target Hardware channel	Active Skill Operation	Triggered Safety Condition	Risk Score (Solver)
1	Bias	Gripper_State	Place	Collision_with_Human	740.35
2	Stuck-At	Camera_Image	Carry to B	Collision_with_Human	676.81
3	Bias	Camera_Image	Carry to B	Collision_with_Human	538.98
4	Bias	Camera_Image	Pick	Collision_with_Human	527.55
5	Noise	Camera_Image	Move to A	Misidentification	244.83
6	Bias	Controller_Bus	Carry to B	Collision_with_Human	151.29

श्री माता वैष्णो देवी विश्वविद्यालय
SHRI MATA VAISHNO DEVI UNIVERSITY

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Date: 10/07/2026

प्रोजेक्ट कार्य प्रमाण-पत्र/Project Work Certificate

यह प्रमाणित किया जाता है कि **श्री विष्णु विश्वकर्मा (अनुक्रमांक-22354)**, जो **अभियांत्रिकी एवं प्रौद्योगिकी संस्थान, डॉ. राम मनोहर लोहिया अवध विश्वविद्यालय, फैजाबाद (अयोध्या) - 224001, उत्तर प्रदेश** के **सूचना प्रौद्योगिकी विभाग** से 4-वर्षीय अवधि वाला B.Tech डिग्री प्रोग्राम (प्रौद्योगिकी में स्नातक की उपाधि) कर रहे हैं, उन्होंने अपना सौंपा गया B.E./B.Tech. अंतिम वर्ष का शैक्षणिक **प्रोजेक्ट कार्य 01.01.2026** से **30.06.2026** तक 6 माह की अवधि हेतु **डॉ. अमित कुमार सिन्हा**, सहायक प्रोफेसर (सुपरवाइज़र/प्रोजेक्ट गाइड) के मार्गदर्शन में **स्कूल ऑफ मैकेनिकल इंजीनियरिंग, अभियांत्रिकी संकाय, श्री माता वैष्णो देवी विश्वविद्यालय, कटड़ा – 182320 (जम्मू व कश्मीर)** में संपन्न किया है।

This is to certify that **Mr. Vishnu Vishwakarma (Roll No. 22354)** pursuing 4-Year duration of B.Tech Degree Program in **Department of Information Technology (IT)** from **Institute of Engineering & Technology (IET), Dr. Ram Manohar Lohia Avadh University (RMLAU), Faizabad (Ayodhya) - 224001, Uttar Pradesh** has carried out his assigned B.E./B.Tech. Final Year Academic **Project Work** under the supervision or guidance of **Dr. Amit Kumar Sinha**, Assistant Professor (Supervisor/Project Guide) at **School of Mechanical Engineering (ME), Faculty of Engineering, Shri Mata Vaishno Devi University (SMVDU), Kakryal, Katra – 182320 (UT of Jammu & Kashmir)** from **01.01.2026** to **30.06.2026** for a period of 6 months.

इस प्रोजेक्ट कार्य की अवधि के दौरान, उन्होंने ***"अवधारणा: ए.आई.-नियंत्रित रोबोटिक प्रणालियों के गतिशील जोखिम आकलन हेतु, रीइन्फोर्समेंट लर्निंग पर आधारित विफलता परिदृश्यों की पहचान (रोबोट्स के प्रशिक्षण और परीक्षण के लिए 3 डी सिमुलेशन)"*** नामक **आई.टी. के डेटा विज्ञान एवं अनुप्रयोग में विशेषज्ञता** पर आधारित तकनीकी प्रोजेक्ट शीर्षक पर कार्य किया जहाँ उन्होंने दो सदस्यों वाली शैक्षणिक प्रोजेक्ट टीम/समूह के सदस्य के तौर पर डायनामिक रिस्क असेसमेंट, हाइब्रिड रिस्क मॉडल, फॉल्ट इंजेक्शन, एम2एम (मॉडल-से-मॉडल) ट्रांसफॉर्मेशन, आर.ओ.एस. (रोबोटिक ऑपरेटिंग सिस्टम), ए.आई.-कंट्रोल्ड रोबोटिक सिस्टम, जेनेटिक एल्गोरिद्म, डीप लर्निंग, लार्ज लैंग्वेज मॉडल (एल.एल.एम.) और रीइन्फोर्समेंट लर्निंग को लागू करने का काम किया है, तथा उन्होंने अपना सौंपा गया प्रोजेक्ट कार्य सफलतापूर्वक पूरा किया। उपरोक्त अवधि के दौरान, उनका चरित्र अत्यंत उत्तम पाया गया।

During this period of project work, he worked on the **I.T. specialization in Data Science and Applications** based technical project titled:- ***"REINFORCEMENT LEARNING-BASED IDENTIFICATION OF FAILURE SCENARIOS FOR CONCEPT: DYNAMIC RISK ASSESSMENT OF AI-CONTROLLED ROBOTIC SYSTEMS (3D Simulation for Training & Testing Robots)"*** where he has performed implementation of the following Dynamic Risk Assessment, Hybrid Risk Models, Fault Injection, M2M (Model-to-Model) Transformation, ROS (Robot Operating System), AI-Controlled Robotic Systems, Genetic Algorithms, Deep Learning, Large Language Models (LLMs) & Reinforcement Learning as a member of a two-member academic project team/group, and he successfully completed his own assigned project work. During the above period, his character was found to be very good.

हम उनके उज्ज्वल भविष्य के लिए उन्हें शुभकामनाएँ देते हैं। We wish him all the best for bright future and endeavours.

Best of Luck! Happy Learning!

10/07/2026

Dr. Amit Kumar Sinha

Assistant Professor (Supervisor)

Department of ME, SMVDU Katra (J&K)

To,

The Principal/HOD of IT,

Institute of Engineering & Technology (RMLAU) Faizabad, Ayodhya

Shri Mata Vaishno Devi University Katra
10/07/2026
Day/Month/Year
School of Mech. Engg.

डॉ0 राम मनोहर लोहिया अवध विश्वविद्यालय, अयोध्या

Thank You!

REFERENCE FROM

विज्ञानं ब्रह्म

श्री माता वैष्णो देवी विश्वविद्यालय, कटड़ा

SHRI MATA VAISHNO DEVI UNIVERSITY, KATRA

THANK YOU!

Project Title: Reinforcement Learning-Based Identification of Failure Scenarios for Concept: Dynamic Risk Assessment of AI-Controlled Robotic Systems (3D Simulation for Training & Testing Robots)

PRESENTED BY

Mr. Vishnu Vishwakarma (22354) & Mr. Gyaneshvar (22328)

B.TECH - INFORMATIOM TECHNOLOGY (IT)

SESSION: 2022 - 2026, JULY 2026

UNDER THE SUPERVISION OF

Er. Akhilesh Kumar, Assistant Professor (Supervisor), Dept. of I.T.

INSTITUTE OF ENGINEERING & TECHNOLOGY (IET)

Dr. Rammanohar Lohia Avadh University (RMLAU), Faizabad, Ayodhya (U.P.)

REFERENCE FROM

SHRI MATA VAISHNO DEVI UNIVERSITY (SMVDU), KATRA (J&K)